Morphology

Lecture-04
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Computational Linguistics

Morphology:

- Scientific study of word formation
- study of patterns of word formation
- How words are orginated, grammatical forms, on what basis the word class is formed.

Word:

A word is a minimal free form of a language that has a given sound, meaning and syntactic function.

Language:

- An arbitrary vocal system used for human communication
- Elements in a language are arranged according to certain rules, but not at random.
- Elements at a lower level are combined according to certain rules to form elements at a higher level.

Lexicology:

- A branch of linguistics concerned with the study of the vocabulary of a given language.
- Deals with words, their origin, development, structure, formation, meaning and usage

Morpheme:

- Minimal, meaningful unit of a word or in the grammar of a language.
- Not further divisible or analyzable into smaller forms.
- Breaking up morpheme loses its identity and leaves its meaningless noises

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- May be a word or not a word
- A synonym for morpheme is glosseme

Example:

Unladylike ⟶ Un + lady + like

Encouragement ⟶ En + courage + ment

Disillusionment ⟶ Dis + illusion + ment

Morph:

- A morph is the phonetic realization of a morpheme.
- The actual utterance of the morpheme.

Allomorph:

- Allomorph is phonologically distinct variants of the same morpheme
- Different realization, manifestation of the same morpheme.

Examples of Allomorph:

[s] ⟶ English plural morpheme

[-s]

cat	thank
cats	thanks

Here [-s] with morphs ending in /p, t, k/

[-z]

dog	rim
dogs	rims
↓	↓
dogz	rimz

Here [-z] with morphs ending in /b, d, g, v, l/

[-iz]

bus	horse
buses	horses

Here [-iz] with morphs ending in /z, t/

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Morpheme types

- Free morpheme

(A free morpheme is a word that can occur by itself)

(It can stand alone)

(We can add other morphemes in it)

Example: boy, man, establish are free morphemes

- Bound morpheme

(A grammatical unit that never occurs by itself, but is always attached to some other morphemes)

Examples:

Bound Morpheme	Words
[s]	cats
[ness]	kindness
[ize]	visualize

Affixation

An affix is a bound morpheme that is joined before, after or within a root or stem

Types of affixes

- Prefix

(Added before the root or stem of the word)

Example: Unpredictable, Unresponsive, reschedule, etc

- Suffix

(Attached to the end of a root or stem)

Example: cats, eaten, eating, higher, highest

- Infix

(Inserted within a root or stem)

(Added in the base form of a word to create a new word or intensify meaning)

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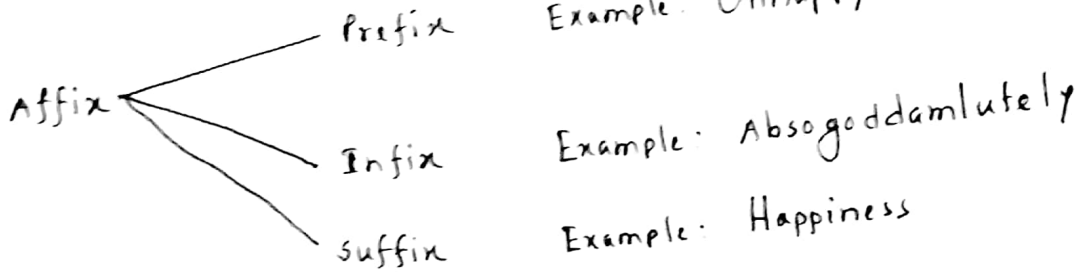
Example:

Absogoddam lutely



- Circumfixes:

Consists of two parts

- A prefix and a suffix that together create a new word

Example:

In Dutch 'berg' means 'mountain'

ge-berg-te means mountain chain

- Simulfix:

A simulfix is a change or replacement of vowels or consonants (usually vowels) which changes the meaning of a word

Example:

eat ⟶ ate

tooth ⟶ teeth

Morphemes and classes

- Root

(A root is the irreducible core of a word, with absolutely nothing else attached to it)

Example:

jump – jumps
jump – jumping
jump – jumped

} Here, jump is the root

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stem

A stem deals with inflectional morphology
(Inflectional only, but not derivational affixes added to it)

Example:

friend ⟶ friends
friendship ⟶ friendships
greenhouse ⟶ greenhouses

worker ⟶ workers
shift ⟶ shifted

The stem is that part of a word that exists before the addition of any inflectional morpheme.

- Base

A base is any form to which affixes of any kind can be added.
It can be both inflectional or derivational

boy ⟶ boys
boy ⟶ boyish
boy ⟶ boyhood

[All roots are bases, bases are called stem in context of inflectional morphology]

Bound Morpheme

Definition: must be attached to another morpheme

Bound Morpheme

Derivational morpheme
a. may change syntactic class
b. to form new words
c. examples: -able, un-, re-, etc

Inflectional morpheme
a. Different forms of the same word
b. Not change syntactic class
c. Only 8 kinds in English:
-'s, -s (plural nouns), -ing, -ed/-e
-est, -er, -s (SV agreement)

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Formation of new words:

1. **Derivational Morpheme**

- Adding an affix to the root or stem
- changing the meaning of the base to which they are attached

a) changing the meaning of the base to which they are attached

kind ⟶ unkind (Adjectives)

obey ⟶ disobey (Verbs)

b) by changing the word-class that a base belongs

kind ⟶ kindly

simple ⟶ simply

Derivation of Nouns:

Verb ⟶ Noun

speak ⟶ Speaker

swim ⟶ Swimming

read ⟶ read / reading

Adjective ⟶ Noun

kind ⟶ kindness

good ⟶ goodness

black ⟶ blackness

Noun ⟶ Noun

mother ⟶ motherhood

sense ⟶ non-sense

Derivation of Adjective:

Noun ⟶ Adjective

master ⟶ masterly

brother ⟶ brotherly

Verb ⟶ adjective

read ⟶ readable

love ⟶ loveable

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jective ⟶ Adjective

common ⟶ uncommon

possible ⟶ impossible

Derivation of Verbs

Noun ⟶ Verb

analysis ⟶ analyze

slave ⟶ enslave

Adjective ⟶ Verb

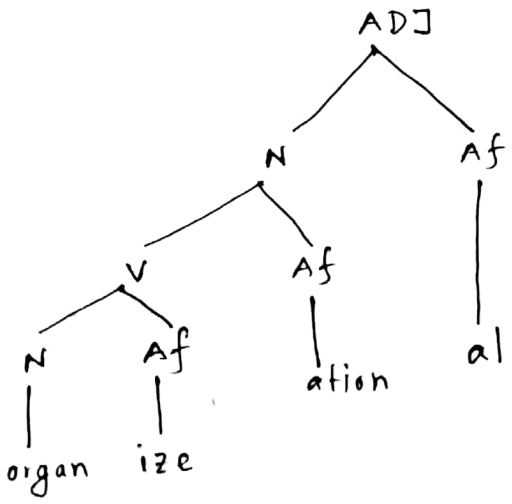
rich ⟶ enrich

legal ⟶ legalize

Verb ⟶ verb

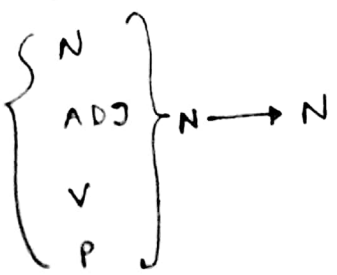
continue ⟶ discontinue

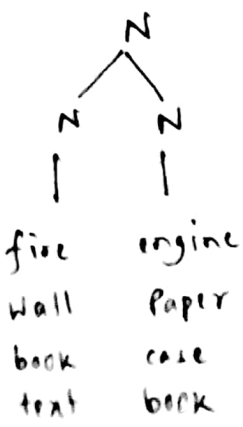
pack ⟶ Unpack

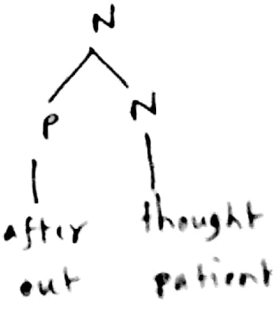


Compounding

When two words or at least two root morphemes are joined together, the derived word results a compounding.







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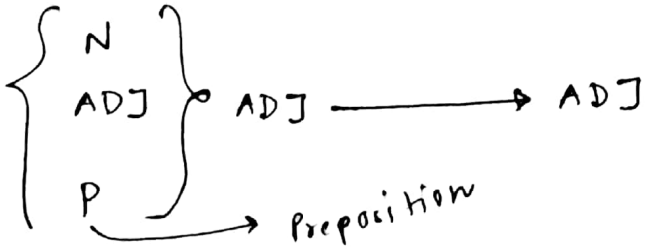

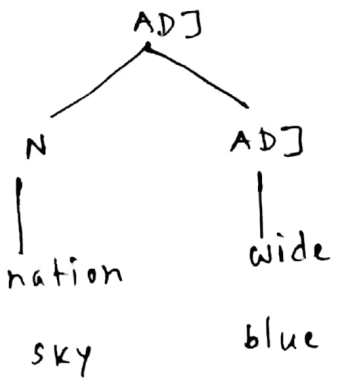

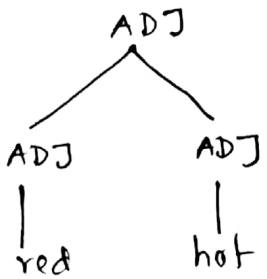

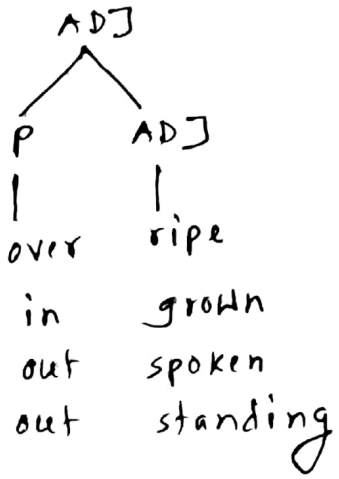

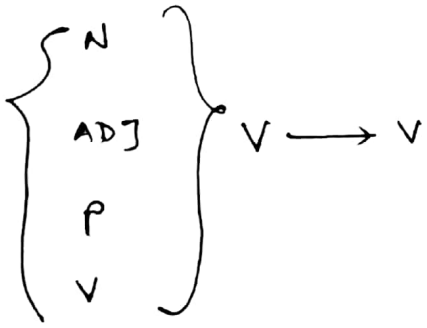

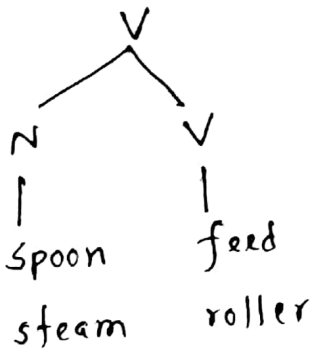

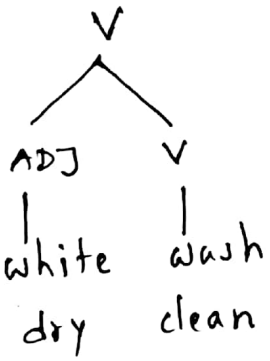

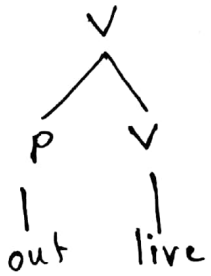

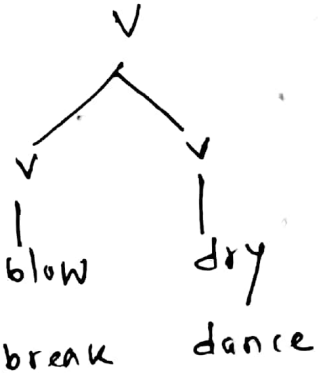

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Duplication

- In this process all or part of the base is repeated as prefix or suffix
- Full or partial repetition of a free morpheme; sometimes with variation

Full	With variation
so-so	zigzag
bye-bye	dilly-dally
	hotch potch
	hodge podge
	mishmash
	willy-nilly
	hanky-panky

Blending:

Similar to compounding, but parts of the free morphemes involved are lost
(Usually 1st part of 1st word + end of 2nd word)

Word 1	Word 2	Word
breakfast	lunch	brunch
smoke	fog	smog
motor	hotel	motel
news	broadcast	newscast

Clipping (clipped forms)

Part of a free morpheme is cut off (i.e., shortening a polysyllabic word); often in casual speech.

Professor ⟶ Prof.
advertisement ⟶ ad
document ⟶ doc
Laboratory ⟶ lab
Automatic ⟶ auto

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Acronyms:

Abbreviate a longer term by taking the initial letters

Example:

NATO (North Atlantic Treaty Organization)
TOEFL (Test of English as a Foreign Language)
AIDS (Acquired Immune Deficiency Syndrome)
NASA (National Aeronautics & Space Administration)

If unpronounceable ⟶ each letter is sounded out separately

ATM (Automatic teller machine)
VCR (Video cassette Recorder)

Back formation

A word (usually a noun) is reduced to form another word of a different type (usually a verb).

editor ⟶ edit
donation ⟶ donate
babysitter ⟶ babysit
terminator ⟶ terminate

[The direction of derivation is inverted]

먹기 ⟶ 먹

Coining

Creating a completely new free morpheme, which is unrelated to any existing morphemes; a rare thing

Example: googol
Nylon

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Conversion

In some languages it is possible to form new lexeme merely by shifting the word category without adding any affixes.

Example:

table ⟶ to table
chair ⟶ to chair
water ⟶ to water

Onomatopoeia:

Words imitate sounds in nature

Example:

A cat ⟶ meow
A cow ⟶ moo
Ring of a bell → ding-dong

Extension of word formation rules

hamburger
cheeseburger
fishburger

marathon
telethon

alcoholic
workaholic

[part of a word is treated as a morpheme though it's not]

Synthetic Compound

The head lexeme is derived from a verb

Example:

hand-washing
User-driven
home-made

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Portmanteau

When a single morpheme represents simultaneously two or more grammatical functions is called portmanteau morpheme.

Example:

[-s] is used to denote 3rd person singular number
[-s] is used to denote plural number

* Run ⟶ Run ⟶ Run
  cut ⟶ cut ⟶ cut

A Single Word with different morphemes

- One morpheme – nation
- Two morphemes – national
- Three morphemes – national -ize
- Four morphemes – de nation al ize
- More than four morphemes – de nation al iz ation

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LLM = Large Language Model

A Large Language Model **is a type of** artificial intelligence (AI) **model designed to** understand, generate, and process human language. It is a type of **machine learning model**, specifically based on **deep learning** using **neural networks**.

Working principle:

- The model is trained on **massive text datasets** (books, articles, websites, etc.).
- It learns **patterns in language**, such as grammar, meaning, and context.
- Once trained, it can **generate human-like text**, **answer questions**, **summarize**, **translate**, **analyze sentiment**, and more.

Examples of LLMs:

- **GPT-5** (by OpenAI)
- **Gemini** (by Google DeepMind)
- **Claude** (by Anthropic)
- **LLaMA** (by Meta)

Mathematical overview:

An LLM is a parameterized probability model that estimates

$$P(w_t / w_1, w_2, \ldots, w_{t-1})$$

This means it predicts the next word in a sequence given all previous words.

Objective of an LLM
An **LLM** is a probabilistic model that tries to estimate the likelihood of a sequence of words:
$P(w_1, w_2, \ldots, w_T)$

where w_i is the i^{th} word (or token) in the sequence.
By the chain rule of probability:

$$P(w_1, w_2, \ldots, w_T) = \prod_{t=1}^{T} P(w_t \mid w_1, w_2, \ldots, w_{t-1})$$

So the model's goal is to predict the next token given all previous tokens.

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Problem:

Show the full mathematical flow of a mini LLM with 1 layer, 3 tokens, and tiny dimensions (2D) for the following sentence fragment to predict the next word:

"The cat sat"

Solution:

Given that,

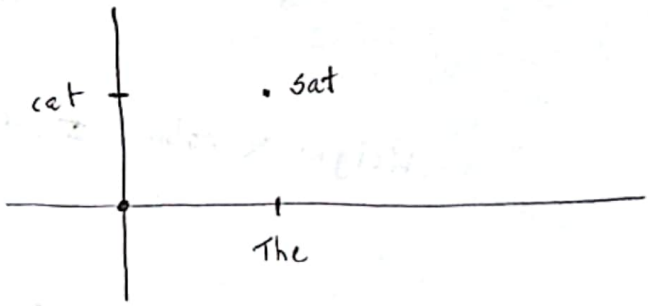
- Number of tokens (T) = 3
- Number of layers (L) = 1

Let us consider

- Embedding dimension (d) = 2
- Attention dimension (d_k) = 2

Step 1: Token Embedding

Token	Embedding x_t
The	[1, 0]
cat	[0, 1]
sat	[1, 1]



The input matrix is:

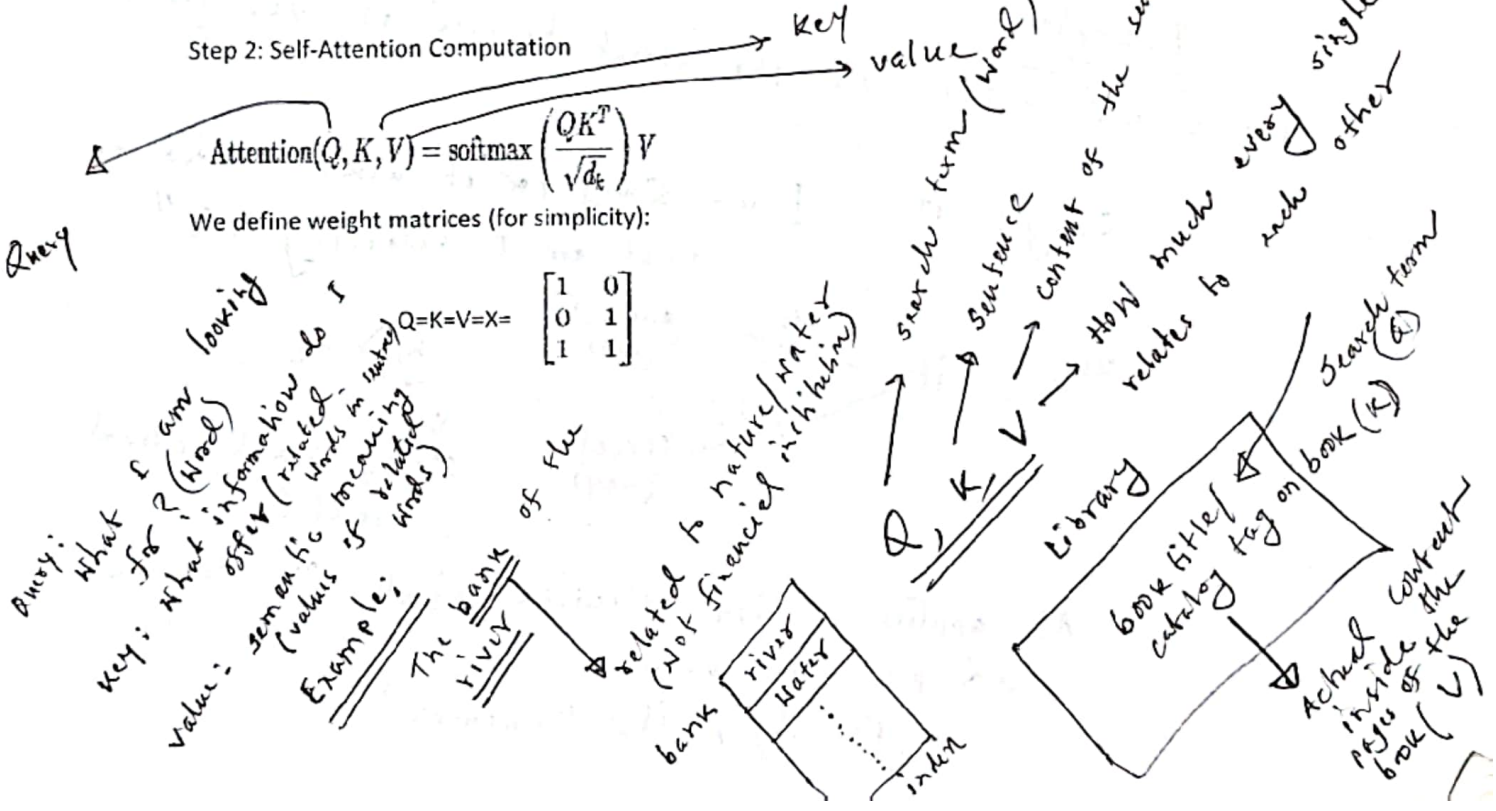
$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Step 2: Self-Attention Computation

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

We define weight matrices (for simplicity):

$$Q=K=V=X=\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$



$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \quad K^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

(a) Compute QK^T

3×2 · 2×3 → 3×3

$$QK^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Then divide by $\sqrt{d_k} = \sqrt{2} \approx 1.414$: (Attention dimension)

$$\frac{QK^T}{\sqrt{2}} = \begin{bmatrix} 0.71 & 0 & 0.71 \\ 0 & 0.71 & 0.71 \\ 0.71 & 0.71 & 1.41 \end{bmatrix}$$

(b) Apply Softmax row-wise

For each row, we apply

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

For row 1: [0.71, 0, 0.71]

Exponentiation → [2.03, 1, 2.03]

Normalization → [0.41, 0.18, 0.41]

Similarly, applying softmax, we obtain the following:

Token	Softmax weights (attention to [The, cat, sat])
The	[0.41, 0.18, 0.41]
cat	[0.18, 0.41, 0.41]
sat	[0.28, 0.28, 0.44]

(c) Compute Attention Output

Attention (Q, K, V) =weights × V

Since V=X, we multiply each attention row by X:

1. For "The":

$$[0.41, 0.18, 0.41] \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = [0.82, 0.59]$$

2. For "cat":

$$[0.18, 0.41, 0.41]X = [0.59, 0.82]$$

3. For "sat":

$$[0.28, 0.28, 0.44]X = [0.72, 0.72]$$

Thus attention outputs:

$$A = \begin{bmatrix} 0.82 & 0.59 \\ 0.59 & 0.82 \\ 0.72 & 0.72 \end{bmatrix}$$

Step 3: Feed-Forward Layer

Each token vector passes through a simple fully connected feed-forward layer:

$$h_t = \mathbf{ReLU}(A_t W_1 + b_1)$$

Let

$$W_1 = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad b_1 = [0, 0]$$

For "The":

$$h_1 = [0.82, 0.59] \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

= [1.41, -0.23]=> ReLU(=[1.41, -0.23])= =[1.41, 0]

For "cat":

$$h_2 = [0.59, 0.82] \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

= [1.41, 0.23]=> ReLU(=[1.41, 0.23])= =[1.41, 0.23]

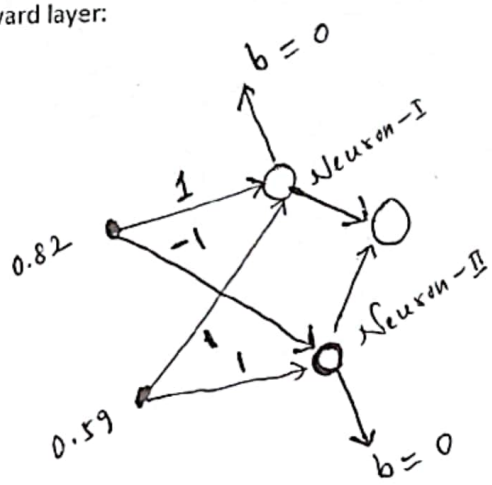
For "sat":

$$h_3 = [0.72, 0.72] \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

= [1.44, 0]=> ReLU(=[1.44, 0])= =[1.44, 0]

$$H = \begin{bmatrix} 1.41 & 0 \\ 1.41 & 0.23 \\ 1.44 & 0 \end{bmatrix} \begin{matrix} \leftarrow h_1 \\ \leftarrow h_2 \\ \leftarrow h_3 \end{matrix}$$

sat



Neuron - I

$(0.82 * 1 + 0.59 * 1) + b$

$= 1.41 + b = 1.41$

$(0.82 * (-1) + 0.59 * 1)$

Neuron-II

$+ b$

$= -0.23 + 0$

$= -0.23$

Relu = $\begin{cases} \max(0, x) = 1.41 & x > 0 \\ 0 & x < 0 \\ -0.23 & \end{cases}$

Step 4: Output Probability (Softmax over Vocabulary)

Assume a tiny vocabulary: {on, dog, under}.

Consider the output weight matrix

$$W_0=\begin{bmatrix}1 & 0\\ 0 & 1\\ 1 & 1\end{bmatrix}$$ → Assumed

For the last token "sat"

$[1.44, \quad 0]$ ← h_1

$$\text{logits}=\left[W_0 \times h_t^T\right]^T$$

$$W_0 h_t^T=\begin{bmatrix}1 & 0\\ 0 & 1\\ 1 & 1\end{bmatrix} \times [1.44, \quad 0]^T= \begin{bmatrix}1 & 0\\ 0 & 1\\ 1 & 1\end{bmatrix} \times \begin{bmatrix}1.44\\ 0\end{bmatrix}= \begin{bmatrix}1.44\\ 0\\ 1.44\end{bmatrix}$$

Logits = [1.44, 0, 1.44]

Apply softmax:

softmax ([1.44, 0, 1.44])

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

$z_i=1.44$; $e^{zi}=4.22$; $z_i=0$; $e^{zi}=1$; $z_i = 1.44$; $e^{z_i} = 4.22$

Summation= 4.22+1+4.22=9.44

softmax([1.44, 0, 1.44])= [0.45, 0.11, 0.45]

$$\frac{4.22}{4.22+1+4.22} = 0.45 \qquad \frac{1}{4.22+1+4.22} = 0.11 \qquad \frac{4.22}{4.22+1+4.22} = 0.45$$

Interpretation:
The model predicts the next word with:

- P("on") = 0.45
- P("dog") = 0.11
- P("under") = 0.45

highest probability

So it's most likely to choose "on" or "under" as the next token.

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ReLU:

The ReLU function can be described mathematically as follows:

$$f(x) = \max(0, x)$$

Where:

- x is the input to the neuron.
- The function returns x if x is greater than 0.
- If x is less than or equal to 0, the function returns 0.

The formula can also be written as:

$$f(x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

This simplicity is what makes ReLU so effective in training deep neural networks, as it helps to maintain non-linearity without complicated transformations, allowing models to learn more efficiently.

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K- Nearest Neighbor Algorithm

Find a phoneme for test frame using the Euclidean distance measure.

/a/ :	2	4	5	Training Data
/a/ :	3	5	6	
/i/ :	4	5	3	
/i/ :	3	4	5	
/i/ :	2	3	4	

Test Data : 2 5 4

Answer:

From Training Data :

$dist_a^2 = (2-2)^2 + (4-5)^2 + (5-4)^2 = 2$ *

$dist_a^2 = (3-2)^2 + (5-5)^2 + (6-4)^2 = 5$

$dist_i^2 = (4-2)^2 + (5-5)^2 + (3-4)^2 = 5$

$dist_i^2 = (3-2)^2 + (4-5)^2 + (5-4)^2 = 3$ *

$dist_i^2 = (2-2)^2 + (3-5)^2 + (4-4)^2 = 4$ *

if $K = 3$,

Minimum are $dist_a^2$, $dist_i^2$, $dist_i^2$

If we take majority voting,

One vote for a

Two vote for i

Winner is i

Output phoneme = /i/

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Nearest Neighbor Algorithm

Find a phoneme for test frame using the Euclidean distance measure.

/a/ :	2	4	5	Training data
/a/ :	3	5	6	
/i/ :	4	5	3	
/i/ :	3	4	5	
/i/ :	2	3	4	
Test data :	2	5	4	

Answer:

Training Data:

For /a/ : Mean calculation

$$\mu_{1a} = \frac{2+3}{2} = 2.5, \quad \mu_{2a} = \frac{4+5}{2} = 4.5, \quad \mu_{3a} = \frac{5+6}{2} = 5.5$$

For /i/ : Mean calculation

$$\mu_{1i} = \frac{4+3+2}{3} = 3, \quad \mu_{2i} = \frac{5+4+3}{3} = 4, \quad \mu_{3i} = \frac{3+5+4}{3} = 4$$

Testing Data:

Using Euclidean distance measure

$$d_a^2 = (\mu_{1a} - 2)^2 + (\mu_{2a} - 5)^2 + (\mu_{3a} - 4)^2$$
$$= (2.5-2)^2 + (4.5-5)^2 + (5.5-4)^2 = 2.75$$

$$d_i^2 = (\mu_{1i} - 2)^2 + (\mu_{2i} - 5)^2 + (\mu_{3i} - 4)^2$$
$$= (3-2)^2 + (4-5)^2 + (4-4)^2 = 2$$

Since $d_i^2 < d_a^2$, output phoneme = /i/

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Nearest Neighbor Algorithm

Find a phoneme for test frame using the ~~Euclidean distance~~ Mahalanobis distance measure.

Training Data:

/a/:	2	1	/i/:	3	2	/u/:	5	3
/a/:	1	2	/i/:	1	3	/u/:	4	2

Test Data: 2 3

Ans:

For /a/ phoneme:

Mean calculation:

$$\mu_{1a} = \frac{2+1}{2} = 1.5$$

$$\mu_{2a} = \frac{1+2}{2} = 1.5$$

Covariance calculation:

$$\Sigma_{1,a} = \frac{(1.5-2)^2 + (1.5-1)^2}{2} = 0.25$$

$$\Sigma_{2,a} = \frac{(1.5-1)^2 + (1.5-2)^2}{2} = 0.25$$

~~Inverse~~ Covariance Matrix:

$$\Sigma_a = \begin{bmatrix} \Sigma_{1,a} & 0 \\ 0 & \Sigma_{2,a} \end{bmatrix}$$

$$= \begin{bmatrix} 0.25 & 0 \\ 0 & 0.25 \end{bmatrix}$$

Inverse covariance matrix:

$$\Sigma_a^{-1} = \begin{bmatrix} \frac{1}{0.25} & 0 \\ 0 & \frac{1}{0.25} \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

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For /i/ phoneme:

Mean calculation:

$$\mu_{1,i} = \frac{3+1}{2} = 2$$

$$\mu_{2,i} = \frac{2+3}{2} = 2.5$$

Covariance calculation:

$$\Sigma_{1,i} = \frac{(2-3)^2 + (2-1)^2}{2} = 1$$

$$\Sigma_{2,i} = \frac{(2.5-2)^2 + (2.5-3)^2}{2} = 0.25$$

Diagonal covariance Matrix:

$$\Sigma_i = \begin{bmatrix} 1 & 0 \\ 0 & 0.25 \end{bmatrix}$$

Inverse covariance Matrix:

$$\Sigma_i^{-1} = \begin{bmatrix} \frac{1}{1} & 0 \\ 0 & \frac{1}{0.25} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

For /u/ Phoneme

Mean calculation:

$$\mu_{1,u} = \frac{5+4}{2} = 4.5$$

$$\mu_{2,u} = \frac{3+2}{2} = 2.5$$

Covariance calculation:

$$\Sigma_{1,u} = \frac{(4.5-5)^2 + (4.5-4)^2}{2} = 0.25$$

$$\Sigma_{2,u} = \frac{(2.5-3)^2 + (2.5-2)^2}{2} = 0.25$$

Scanned with CamScanner

Diagonal covariance Matrix:

$$\Sigma_u = \begin{bmatrix} 0.25 & 0 \\ 0 & 0.25 \end{bmatrix}$$

Inverse covariance Matrix:

$$\Sigma_u^{-1} = \begin{bmatrix} \frac{1}{0.25} & 0 \\ 0 & \frac{1}{0.25} \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

Mahalanobis Distance Calculation:

$$x = \begin{bmatrix} 2 & 3 \end{bmatrix}$$

$$d_a^2 = (x - \mu_a) \Sigma_a^{-1} (x - \mu_a)^T$$

$$= \left(\begin{bmatrix} 2 & 3 \end{bmatrix} - \begin{bmatrix} 1.5 & 1.5 \end{bmatrix} \right) \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \left(\begin{bmatrix} 2 & 3 \end{bmatrix} - \begin{bmatrix} 1.5 & 1.5 \end{bmatrix} \right)^T$$

$$= \begin{bmatrix} 0.5 & 1.5 \end{bmatrix} \times \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \times \begin{bmatrix} 0.5 & 1.5 \end{bmatrix}^T$$

$$= \begin{bmatrix} 0.5 & 1.5 \end{bmatrix} \times \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \times \begin{bmatrix} 0.5 \\ 1.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 1.5 \end{bmatrix} \times \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

$$= 1 + 9$$

$$= 10$$

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(6)

$$d_i^r = \left(\begin{bmatrix} 2 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 2.5 \end{bmatrix} \right) \times \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \times \left(\begin{bmatrix} 2 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 2.5 \end{bmatrix} \right)^T$$

$$= \begin{bmatrix} 0 & 0.5 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0.5 \end{bmatrix} \times \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$= 0 + 1$$

$$= 1$$

$$d_u^r = \left(\begin{bmatrix} 2 & 3 \end{bmatrix} - \begin{bmatrix} 4.5 & 2.5 \end{bmatrix} \right) \times \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \times \left(\begin{bmatrix} 2 & 3 \end{bmatrix} - \begin{bmatrix} 4.5 & 2.5 \end{bmatrix} \right)^T$$

$$= \begin{bmatrix} -2.5 & 0.5 \end{bmatrix} \times \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \times \begin{bmatrix} -2.5 \\ 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} -2.5 & 0.5 \end{bmatrix} \times \begin{bmatrix} -10 \\ 2 \end{bmatrix}$$

$$= 25 + 1$$

$$= 26$$

Here, $d_a^r = 10$

$d_i^r = 1$

$d_u^r = 26$

phoneme = $\text{argmin}(d_a^r, d_i^r, d_u^r)$

= /i/

Output phone = /i/

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Hidden Markov Model (HMM)

Hidden Markov Model needs

- Initial probability
- Transition probability
- Observation probability

Example:

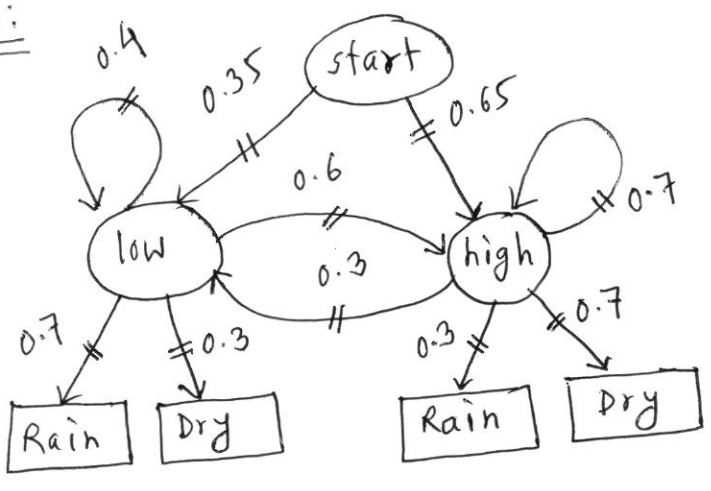
For forecasting weather, it is observed that low atmospheric pressure causes a chance of 70% Rain, otherwise Dry. On the other hand, 70% chance of Dryness is happened if high atmospheric pressure prevails. Again, there is a 60% probability of changing from low to high atmospheric pressure, but the high to low changing occurs in a probability of 30%. The initial probability of high atmospheric pressure is 65%.

Answer the following questions

a) Draw a hidden Markov model (HMM)

b) If rainy weather prevails on today, take a decision whether the cricket match on ~~the day immediate after~~ tomorrow will be happened or not.

a) Ans:



Initial probability:

$$P(\text{high} \mid \text{start}) = 0.65$$
$$P(\text{low} \mid \text{start}) = 0.35$$

Transition probability:

	low	high
low	0.4	0.6
high	0.3	0.7

$$P(\text{low} \mid \text{low}) = 0.4, \quad P(\text{high} \mid \text{low}) = 0.6$$
$$P(\text{low} \mid \text{high}) = 0.3, \quad P(\text{high} \mid \text{high}) = 0.7$$

observation probability:

$$P(\text{Rain} \mid \text{low}) = 0.7 \quad | \quad P(\text{Rain} \mid \text{high}) = 0.3$$
$$P(\text{Dry} \mid \text{low}) = 0.3 \quad | \quad P(\text{Dry} \mid \text{high}) = 0.7$$

b) Ans:

$$P(\text{Rain Rain}) = ?$$
$$P(\text{Rain Dry}) = ?$$

For P(Rain Rain)

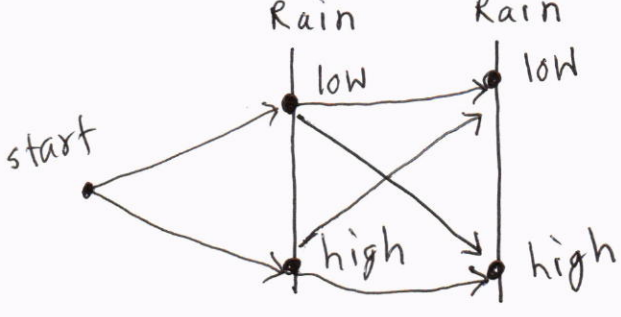


Figure 1

For P(Rain Dry)

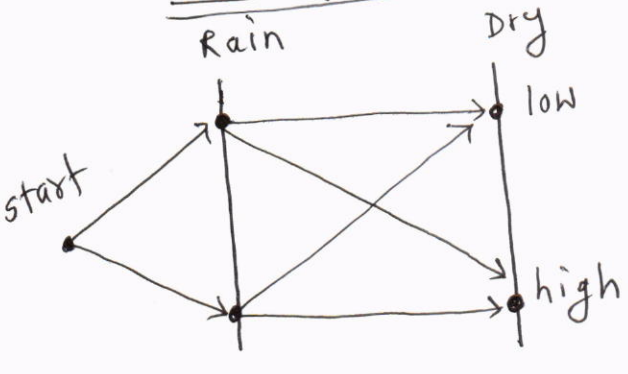


Figure 2

For Figure 1

Hidden state Sequence	Path Probabilities
start → low → low	P(low|start) * P(Rain|low) * P(low|low) * P(Rain|low) = 0.35*0.7*0.4*0.7
start → low → high	P(low|start) * P(Rain|low) * P(high|low) * P(Rain|high) = 0.35*0.7*0.6*0.3
start → high → low	P(high|start) * P(Rain|high) * P(low|high) * P(Rain|low) = 0.65*0.3*0.3*0.7
start → high → high	P(high|start) * P(Rain|high) * P(high|high) * P(Rain|high) = 0.65*0.3*0.7*0.3

Total probability = 0.1946

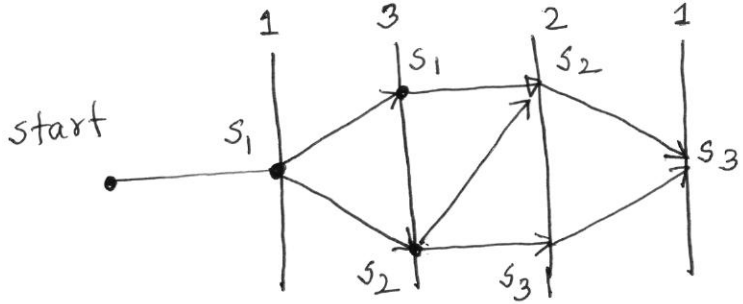
For Figure 2

Hidden state Sequence	Path Probabilities
start → low → low	P(low|start) * P(Rain|low) * P(low|low) * P(Dry|low) = 0.35*0.7*0.4*0.3
start → low → high	P(low|start) * P(Rain|low) * P(high|low) * P(Dry|high) = 0.35*0.7*0.6*0.7
start → high → low	P(high|start) * P(Rain|high) * P(low|high) * P(Dry|low) = 0.65*0.3*0.3*0.3
start → high → high	P(high|start) * P(Rain|high) * P(high|high) * P(Dry|high) = 0.65*0.3*0.7*0.7

Total probability = 0.2454

Since P(Rain Dry) > P(Rain Rain), so the cricket match will be held.

Ans:



HMM for character 'A':

Hidden state sequence	Path Probabilities
$s_1 \to s_1 \to s_2 \to s_3$	$P(s_1\|start) * P(1\|s_1) * P(s_1\|s_1) * P(3\|s_1) * P(s_2\|s_1) * P(2\|s_2) * P(s_3\|s_2) * P(1\|s_3)$ $= 1 * 0.9 * 0.8 * 0 * 0.2 * 0.8 * 0.2 * 0.9 = 0$
$s_1 \to s_2 \to s_2 \to s_3$	$P(s_1\|start) * P(1\|s_1) * P(s_2\|s_1) * P(3\|s_2) * P(s_2\|s_2) * P(2\|s_2) * P(s_3\|s_2) * P(1\|s_3)$ $= 1 * 0.9 * 0.2 * 0.8 * 0.8 * 0.2 * 0.9 = 0.00208$
$s_1 \to s_2 \to s_3 \to s_3$	$P(s_1\|start) * P(1\|s_1) * P(s_2\|s_1) * P(3\|s_2) * P(s_3\|s_2) * P(2\|s_3) * P(s_3\|s_3) * P(1\|s_3)$ $= 1 * 0.9 * 0.2 * 0.1 * 0.2 * 0.1 * 1 * 0.9 = 0.000324$

Total Probability = 0.0023976

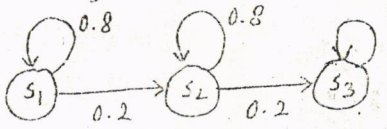
HMM for character 'B':

Hidden state sequence	Path Probabilities
$s_1 \to s_1 \to s_2 \to s_3$	0
$s_1 \to s_2 \to s_2 \to s_3$	0.0027648
$s_1 \to s_2 \to s_3 \to s_3$	0.006912

Total probability = 0.0096768

Therefore, character 'B' is more likely for given observation sequence {1, 3, 2, 1}

The structure of hidden states:



Transition probabilities: $\{a_{ij}\} =$

	S_1	S_2	S_3
S_1	0.8	0.2	0
S_2	0	0.8	0.2
S_3	0	0	1

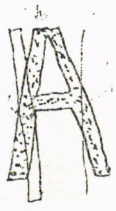
Observation:

observation = number of islands in the vertical slice

For character A:

observation probabilities: $\{b_{jk}\} =$

state \ observation			
S_1	(0.9)	(0.1)	0
S_2	0.1	0.8	0.1
S_3	0.9	0.1	0



For character B:

Observation probabilities:

state \ observation			
S_1	0.9	0.1	0
S_2	0	0.2	0.8
S_3	0.6	0.4	0

Suppose that after character image segmentation the following sequence of island numbers in 4 slices was observed.

{1, 3, 2, 1}

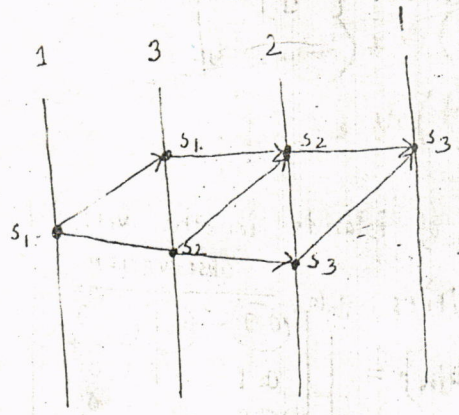
What HMM is more likely to generate this observation sequence, HMM for 'A' or HMM for 'B'?

Ans:

HMM for character 'A':

Hidden state Sequence	Transition Probabilities	Observation Prob	Total
$S_1 \to S_1 \to S_2 \to S_3$	$0.8 * 0.2 * 0.2 *$	$0.9 * 0 * 0.8 * 0.9$	0
$S_1 \to S_2 \to S_2 \to S_3$	$0.2 * 0.8 * 0.2 *$	$0.9 * 0.1 * 0.8 * 0.9$	0.00207
$S_1 \to S_2 \to S_3 \to S_3$	$0.2 * 0.2 * 1 *$	$0.9 * 0.1 * 0.1 * 0.9$	0.000324

Total = 0.0023976



b_{jk} — j: state, k: observation

For A

For calculating state seq 1

$P(S_1) * P(1|S_1) * P(S_1|S_1) * P(3|S_1) * P(S_1|S_1) * P(2|S_2) * P(S_3|S_2) * P(1|S_3)$

$= 1 * b_{11} * P(S_1|S_1) * b_{13} * P(S_2|S_1) * b_{22} * P(S_3|S_2) * b_{31}$

$= P(S_1|S_1) * P(S_2|S_1) * P(S_3|S_2) * b_{11} * b_{13} * b_{22} * b_{31}$

$= 0.8 * 0.2 * 0.2 * 0.9 * 0 * 0.8 * 0.9$

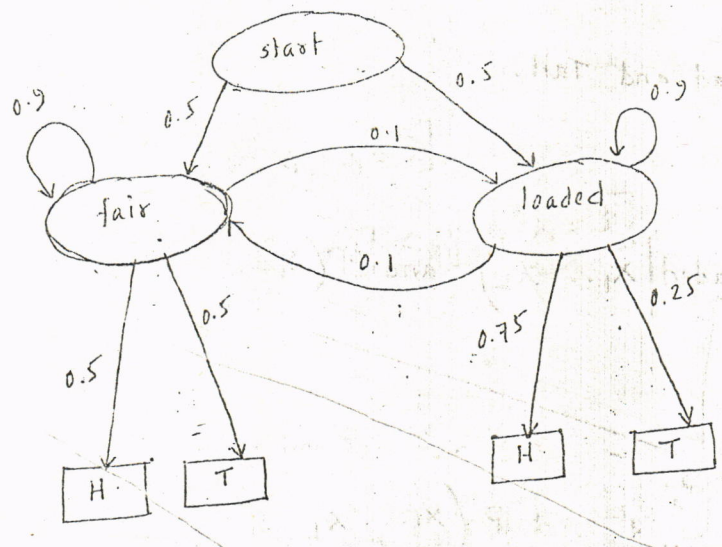
HMM for character 'B'

hidden state Sequence	Transition Probabilities	Observation Prob	Total
$S_1 \to S_1 \to S_2 \to S_3$	$0.8 * 0.2 * 0.2$	$* \; 0.9 * 0 * 0.2 * 0.6$	0
$S_1 \to S_2 \to S_2 \to S_3$	$0.2 * 0.8 * 0.2$	$* \; 0.9 * 0.8 * 0.2 * 0.6$	0.0027648
$S_1 \to S_2 \to S_3 \to S_3$	$0.2 * 0.2 * 1$	$* \; 0.9 * 0.8 * 0.4 * 0.6$	0.006912

Total = 0.0096768

Therefore, character 'B' is more likely for given observation sequence {1, 3, 2, 1}

Coin-Tossing Example :



⇐ Hidden states

⇐ observations

Transition Probabilities:

	fair	loaded
fair	0.9	0.1
loaded	0.1	0.9

Initial Probabilities:

$P(\text{fair}) = 0.5$

$P(\text{loaded}) = 0.9$

Emission or observation Probabilities:

$P(H \mid \text{fair}) = 0.5$ | $P(H \mid \text{loaded}) = 0.75$

$P(T \mid \text{fair}) = 0.5$ | $P(T \mid \text{loaded}) = 0.25$

that are the probabilities for Fair/Loaded coins given the set of tosses

$\{x_1, \dots, x_L\}$ ← Sequence of Head and Tail.

Ans:

$P(s_i = \text{loaded} \mid x_1 \cdots x_L)$ and $P(s_i = \text{Fair} \mid x_1 \cdots x_L)$

s

$$P(s_i = s \mid x_1 \cdots x_L) = \frac{P(x_1 \cdots x_L, s_i = s)}{P(x_1 \cdots x_L)}$$

$$= ???$$

Where $P(x_1 \cdots x_L) = P(x_1 \cdots x_L, s_i = s_1) + P(x_1 \cdots x_L, s_i = s_2) + \cdots + P(x_1 \cdots x_L, s_i = s_L)$

$$= \sum_{s'} P(x_1 \cdots x_L, s_i = s')$$

]

[statistics Formula:

$$P(x_1 \cdots x_L, s_i = s) = P(x_1 \cdots x_i, s_i = s) * P(x_{i+1} \cdots x_L \mid x_1 \cdots x_i, s_i = s)$$

$$= P(x_1 \cdots x_i, s_i = s) * P(x_{i+1} \cdots x_L \mid s_i = s)$$

$f_i(s)$ ← Foward algorithm

$b_i(s)$ ← backward algorithm

Forward algorithm:

Compute $f_i(s) = P[x_1 \cdots x_i, s_i = s]$ for $i = 1 \ldots L$

Base step:

$i=1$, $f_1(s) = P[x_1, s_i = s]$

$= P[s_i = s] * P[x_1 | s_1 = s]$

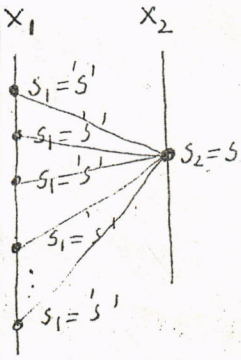
2nd step:

$i=2$, $f_2(s) = P[x_1 \text{ — } x_2, s_2 = s]$

$= P[x_1, s_1 = 's'] * P[s_2 = s | s_1 = 's'] * P[x_2 | s_2 = s]$

$+ P[x_1, s_1 = 's'] * P[s_2 = s | s_1 = 's'] * P[x_2 | s_2 = s]$

$+ \ldots\ldots$



$$= \sum_{s'} \left(P[x_1, s_1 = 's'] * P[s_2 = s | s_1 = 's'] * P[x_2 | s_2 = s] \right)$$

$$= \sum_{s'} \left(f_1(s') * P[s_2 | 's'] * P[x_2 | s] \right)$$

forward probability · Transition probability · observation or emission probability

ith step:

$$f_i(s) = \sum_{s'} \left(f_{i-1}(s') \; P(s|s') * P(x_i|s) \right)$$

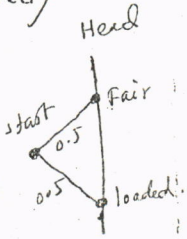
Find $P[S_i = \text{loaded} \mid \text{Head, Head, Tail})$ and $P[S_i = \text{fair} \mid \text{Head, Head, Tail})$ using forward algorithm.

Ans:

Forward algorithm First step:

$$f_1(\text{Loaded}) = P(\text{Head}, S_1 = \text{Loaded}) = P(\text{Loaded}) * P(\text{Head} \mid \text{Loaded})$$
$$= 0.5 * 0.75$$
$$= 0.375$$

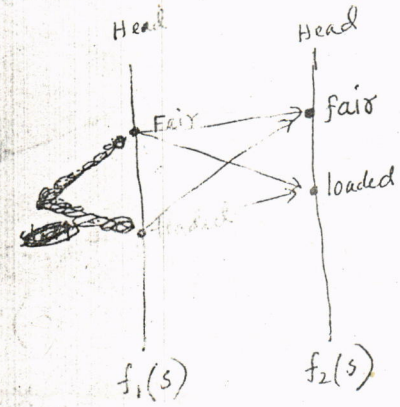
$$f_1(\text{fair}) = P(\text{Head}, S_1 = \text{Fair}) = P(\text{Fair}) * P(\text{Head} \mid \text{Fair})$$
$$= 0.5 * 0.5$$
$$= 0.25$$



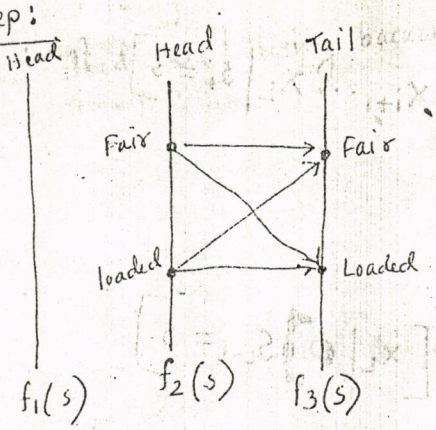
Forward algorithm second step:

$$f_2(\text{Loaded}) = P[X_1 X_2 = \text{Head Head}, S_2 = \text{Loaded}]$$
$$= f_1(\text{Loaded}) * P(\text{loaded} \mid \text{loaded}) * P(\text{Head} \mid \text{loaded}) + f_1(\text{fair}) * P(\text{loaded} \mid \text{fair}) * P(\text{Head} \mid \text{fair})$$
$$= 0.375 * 0.9 * 0.75 + 0.25 * 0.1 * 0.5 = 0.271875$$

$$f_2(\text{fair}) = P[X_1 X_2 = \text{Head Head}, S_2 = \text{fair}]$$
$$= f_1(\text{Loaded}) * P(\text{fair} \mid \text{loaded}) * P(\text{Head} \mid \text{fair}) + f_1(\text{Fair}) * P(\text{Fair} \mid \text{Fair}) * P(\text{Head} \mid \text{Fair})$$
$$= 0.375 * 0.1 * 0.5 + 0.25 * 0.9 * 0.5$$
$$= 0.13125$$



orward algorithm third step:



$$f_3(\text{loaded}) = P[x_1 x_2 x_3 = \text{Head Head Tail},\ s_3 = \text{loaded}]$$

$$= f_2(\text{loaded}) * P[\text{loaded} \mid \text{loaded}] * P[\text{Tail} \mid \text{loaded}]$$
$$+ f_2(\text{fair}) * P[\text{loaded} \mid \text{Fair}] * P[\text{Tail} \mid \text{loaded}]$$

$$= 0.271875 * 0.9 * 0.25 + 0.13125 * 0.1 * 0.25$$

$$= 0.06445$$

$$f_3(\text{Fair}) = P[x_1 x_2 x_3 = \text{Head Head Tail},\ s_3 = \text{Fair}]$$

$$= f_2(\text{loaded}) * P[\text{Fair} \mid \text{loaded}] * P[\text{Tail} \mid \text{Fair}]$$
$$+ f_2(\text{Fair}) * P[\text{Fair} \mid \text{Fair}] * P[\text{Tail} \mid \text{Fair}]$$

$$= 0.271875 * 0.1 * 0.5 + 0.13125 * 0.9 * 0.5$$

Backward algorithm

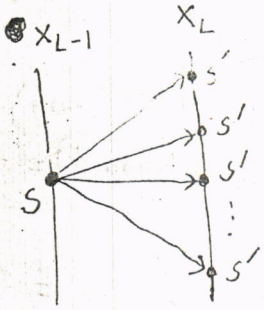
Compute $b_i(s) = P[x_{i+1} \cdots x_L \mid s_i = s]$ for $i = 1 \cdots L$

Base step:

When $i = L-1$,

$$b_{L-1}(s) = P[x_L \mid s_{L-1} = s]$$

$$= P[s' \mid s] * P[x_L \mid s'] + P[s' \mid s] * P[x_L \mid s'] + P[s' \mid s] * P[x_L \mid s'] + \cdots$$



$$= \sum_{s'} P[s' \mid s] * P[x_L \mid s'] = \sum_{s'} \left(P[s' \mid s] * P[x_L \mid s] * b_L(s') \right)$$

($b_L(s') = 1$)

$P[s' \mid s]$: Transition probability

$P[x_L \mid s']$: Emission Probability or observation prob.

$$b_i(s) = \sum_{s'} \left(P[s' \mid s] * P[x_{i+1} \mid s'] * b_{i+1}(s') \right)$$

$P[s' \mid s]$: Transition prob

$P[x_{i+1} \mid s']$: Observation probability

$b_{i+1}(s')$: backward probability.

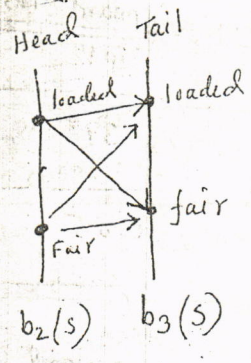
Find $P[s_i = \text{loaded} \mid \text{Head Head Tail}]$ and $P[s_i = \text{fair} \mid \text{Head Head Tail}]$ using backward algorithm.

Ans:

Base step

$$b_3(\text{loaded}) = 1$$

$$b_3(\text{fair}) = 1$$



Backward step-1

$$b_2(\text{loaded}) = P(\text{loaded} \mid \text{loaded}) * P(\text{Tail} \mid \text{loaded}) * b_3(\text{loaded}) + P(\text{fair} \mid \text{loaded}) * P(\text{Tail} \mid \text{fair}) * b_3(\text{fair})$$

$$= 0.9 * 0.25 * 1 + 0.1 * 0.5 * 1$$

$$= 0.275$$

$$b_2(\text{fair}) = P(\text{loaded} \mid \text{Fair}) * P(\text{Tail} \mid \text{loaded}) * b_3(\text{loaded}) + P(\text{fair} \mid \text{fair}) * P(\text{Tail} \mid \text{fair}) * b_3(\text{fair})$$

$$= 0.1 * 0.25 * 1 + 0.9 * 0.5 * 1$$

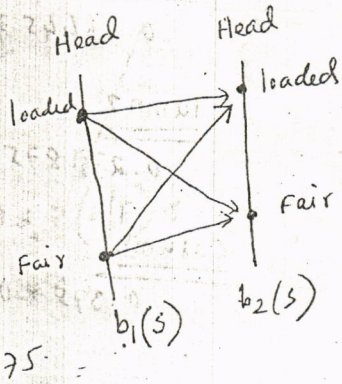
$$= 0.475$$

Backward step-2

$$b_1(\text{loaded}) = P(\text{loaded} \mid \text{loaded}) * P(\text{Head} \mid \text{loaded}) * b_2(\text{loaded}) + P(\text{Fair} \mid \text{loaded}) * P(\text{Head} \mid \text{Fair}) * b_2(\text{fair})$$

$$= 0.9 * 0.75 * 0.275 + 0.1 * 0.5 * 0.475$$

$$= 0.209$$



$$b_1(\text{fair}) = P(\text{loaded} \mid \text{Fair}) * P(\text{Head} \mid \text{loaded}) * b_2(\text{loaded}) + P(\text{fair} \mid \text{Fair}) * P(\text{Head} \mid \text{Fair}) * b_2(\text{Fair})$$

$$= 0.1 * 0.75 * 0.275 + 0.9 * 0.5 * 0.475 = 0.234$$

Ques: Find likelihood of {Head, Head, Tail}

Ans: From Forward Pass:

	step1	step2	step3
loaded	0.375	0.271875	0.06445
Fair	0.25	0.13125	0.07265

From Backward Pass:

	step2	step1	base step
loaded	0.209	0.275	1
Fair	0.234	0.475	1

Likelihood

Column 3

$0.06445 * 1 + 0.07265 * 1 = 0.1371$

Column 2

$0.271875 * 0.275 + 0.13125 * 0.475 = 0.1371$

Column 1

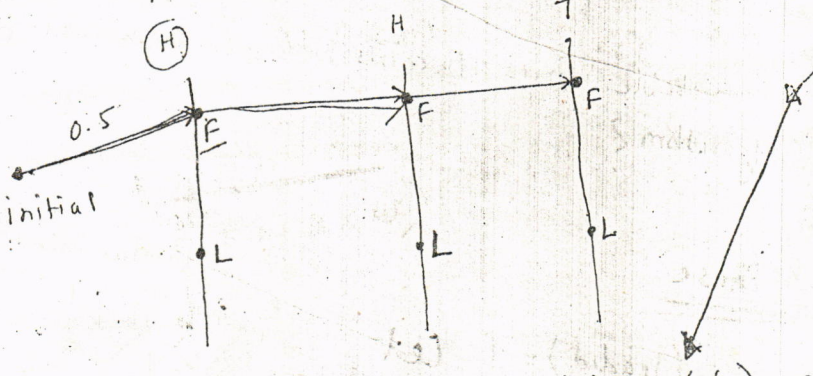
$0.375 * 0.209 + 0.25 * 0.234 = 0.1371$

What is the most probable series of coins to observe sequence {Head, Head, Tail}

Ans:

S_1 S_2 S_3	$P[\text{Head Head Tail}, S_1, S_2, S_3]$
F F F	$(0.5)^3 * 0.5 * (0.9)^2 = 0.050625$
F F L	$(0.5)^2 * 0.25 * 0.5 * 0.9 * 0.1 = 0.0028125$
F L F	$0.5 * 0.75 * 0.5 * 0.5 * 0.1 * 0.1 = 0.0009375$
F L L	$0.5 * 0.75 * 0.25 * 0.5 * 0.1 * 0.9 = 0.00422$
L F F	$0.75 * 0.5 * 0.5 * 0.5 * 0.1 * 0.9 = 0.0084375$
L F L	$0.75 * 0.5 * 0.25 * 0.5 * 0.1 * 0.1 = 0.000468$
L L F	$0.75 * 0.75 * 0.5 * 0.5 * 0.9 * 0.1 = 0.01265$
L L L	$0.75 * 0.75 * 0.25 * 0.5 * 0.9 * 0.9 = 0.0569$ → max

Problem: exponential in length of observation



$0.5 * P(H|F) * P(F|F) * P(H|F) * P(F|F) * P(T|F)$

$= 0.5 * 0.5 * 0.9 * 0.5 * 0.9 * 0.5$

$= (0.5)^3 * 0.5 * (0.9)^2$

Viterbi algorithm: (Backward Starting - Last 2 Start)

Backward phase:

calculate $v_i(s) = P^*[X_{i+1} \dots X_L \mid S_i = s]$

Base:

$$V_L(s) = 1$$

step i:

$$v_i(s) = \max_{s'} \left\{ P(s'|s) * P(X_{i+1}|s') * v_{i+1}(s') \right\}$$

(Transition probability, observation probability, backward probability)

$$\pi_{i+1}(s) = \text{ARGMAX}_{s'} \left\{ P(s'|s) * P(X_{i+1}|s') * v_{i+1}(s') \right\}$$

Forward phase:

Trace path of maximum probability

Base: $\pi_1 = s_1^* = \text{ARGMAX}_{s'} \left\{ P(s') * P(X_1|s') \, v_1(s') \right\}$

step: $s_{i+1}^* = \pi_{i+1}(s_i^*)$

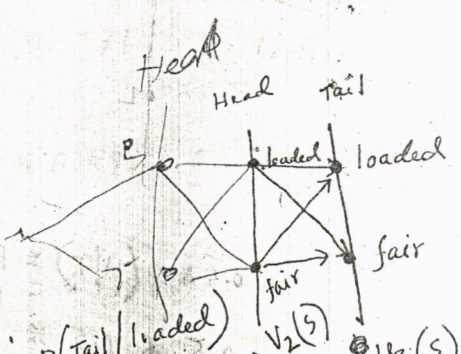
What is the most probable series of coins to observe sequence {Head, Head, Tail} using Viterbi algorithm?

Ans:

Backward Phase

base: $v_3(\text{loaded}) = 1$ let

$v_3(\text{fair}) = 1$



States

0

= 3

= 2

= 1

sequence

step1:

$$v_2(\text{loaded}) = \max \left(P(\text{loaded}|\text{loaded}) * P(\text{Tail}|\text{loaded}) * v_3(\text{loaded}), \; P(\text{fair}|\text{loaded}) * P(\text{Tail}|\text{fair}) * v_3(\text{fair}) \right)$$

$$= \max(0.9 * 0.25 * 1, \; 0.1 * 0.5 * 1)$$

$$= 0.225$$

$\pi_3(\text{loaded}) = \text{loaded}$

$\pi_3(\text{loaded}) =$ Which one is max

$$V_2(\text{Fair}) = \max\Big(P[\text{loaded}|\text{fair}] * P[\text{Tail}|\text{loaded}] * V_3(\text{loaded}),$$
$$P[\text{fair}|\text{fair}] * P[\text{Tail}|\text{fair}] * V_3(\text{fair}) \Big)$$

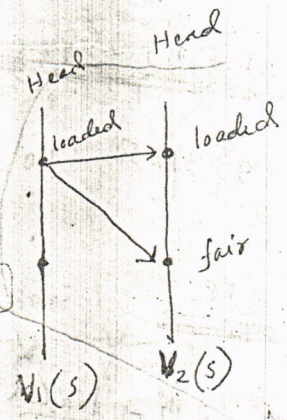
$$= \max(0.1 * 0.25 * 1, \quad 0.9 * 0.5 * 1)$$

$$= 0.45$$

$$\pi_3(\text{Fair}) = \text{Fair}$$

Step 2 :

$$V_1(\text{loaded}) = \max\Big(P[\text{loaded}|\text{loaded}] * P[\text{Head}|\text{loaded}] * V_2(\text{loaded}),$$
$$P[\text{fair}|\text{loaded}] * P[\text{Head}|\text{fair}] * V_2(\text{fair}) \Big)$$

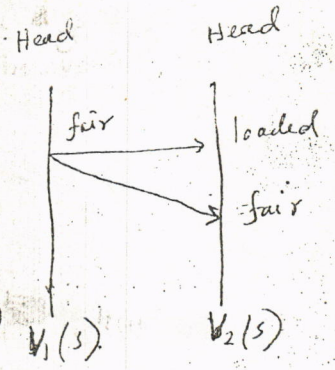


$$= \max(0.9 * 0.75 * 0.225, \quad 0.1 * 0.5 * 0.45)$$

$$= 0.151875$$

$$\pi_2(\text{loaded}) = \text{loaded}$$

$$V_1(\text{fair}) = \max\Big(P[\text{loaded}|\text{fair}] * P(\text{Head}|\text{loaded}) * V_2(\text{loaded}),$$
$$P[\text{fair}|\text{fair}] * P(\text{Head}|\text{fair}) * V_2(\text{fair}) \Big)$$

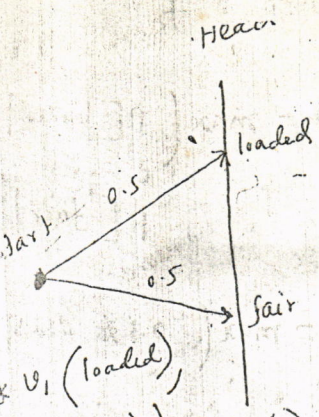


$$= \max(0.1 * 0.75 * 0.225, \quad 0.9 * 0.5 * 0.45)$$

$$= 0.2025$$

$$\pi_2(\text{Fair}) = \text{Fair}$$

step3:



P(Head Head Tail)

= ~~max(P(loaded) * P(loaded)~~

= max(P(loaded) * P(Head | loaded) * V_1(loaded),
P(fair) * P(Head | fair) * V_1(fair))

= max(0.5 * 0.75 * 0.875, 0.5 * 0.5 * 0.2025)

= 0.0569 : 3125

Forward phase

$\pi_1^* = s_1^*$ = loaded

$s_2^* = \pi_2(s_1^*)$

= π_2(loaded)

= loaded

$s_3^* = \pi_3(s_2^*)$

= π_3(loaded)

= loaded

Most Probable coin sequence = $\{s_1^* s_2^* s_3^*\}$

= {loaded loaded loaded}

π_3(head) = L

π_3(F) = F

π_2(L) = L

π_2(F) = F

π_1(L) = L

$\pi_1 = s_1^*$ = F

$s_2^* = \pi_2$(F)

= L

$s_3^* = \pi_3$(L)

= F

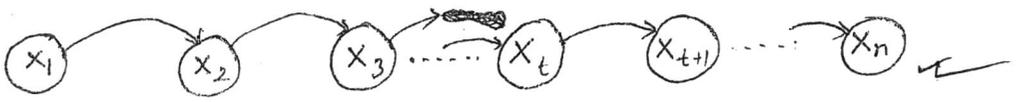
F L F L F

Dr. Md. Nurul Huda
Markov Model
MSCSE, Advanced AI

Markov Process:

Markov Property:

The state of the system at time t+1 depends only on the state of the system at time t.



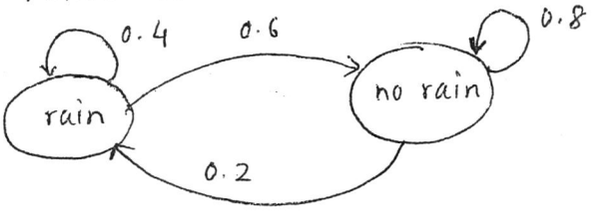
Example1

Weather:

Raining today → 40% rain tomorrow
Raining today → 60% no rain tomorrow

Not raining today → 20% rain tomorrow
Not raining today → 80% no rain tomorrow

Stochastic Finite State Machine (FSM):



The transition matrix:

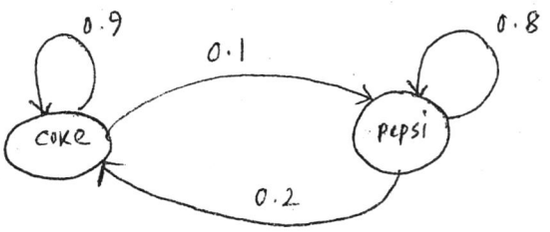
$$P = \begin{array}{c} \\ \text{Rain} \\ \text{No Rain} \end{array} \begin{array}{c} \begin{array}{cc} \text{Rain} & \text{No Rain} \end{array} \\ \begin{bmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{bmatrix} \end{array}$$

↑ stochastic matrix since rows sum up to 1.

[Rows and columns sum up to 1]

Example 2:

- Given that a person's last cola purchase was coke, there is a 90% chance that his next cola purchase will also be coke.
- If a person's last cola purchase was pepsi, there is an 80% chance that his next ~~data~~ cola purchase will also be pepsi.



Transition matrix:

$$P = \begin{array}{c} \\ \text{coke} \\ \text{pepsi} \end{array} \begin{array}{c} \begin{array}{cc} \text{coke} & \text{pepsi} \end{array} \\ \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \end{array}$$

Problem1:

Given that a person is currently a pepsi purchaser, what is the probability that he will purchase coke two purchases from now

Ans:

Method1: $P[\text{Pepsi} \longrightarrow ? \longrightarrow \text{coke}] = P[\text{Pepsi} \rightarrow \text{coke} \rightarrow \text{coke}] + P[\text{Pepsi} \rightarrow \text{Pepsi} \rightarrow \text{coke}]$

$$= P[\text{coke}|\text{coke}] * P[\text{coke}|\text{Pepsi}] + P[\text{coke}|\text{Pepsi}] * P[\text{Pepsi}|\text{Pepsi}]$$

$$= 0.9 * 0.2 + 0.2 * 0.8$$

$$= 0.34 \quad \text{Ans:}$$

Method2:

$$P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

$$P = P \times P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

$$= \begin{array}{c} \\ \text{coke} \\ \text{pepsi} \end{array} \begin{array}{c} \begin{array}{cc} \text{coke} & \text{pepsi} \end{array} \\ \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix} \end{array}$$

$$P[\text{Pepsi} \longrightarrow 2 \longrightarrow \text{coke}] = 0.34$$

Problem 2 :

Given that a person is currently a coke purchaser, what is the probability that he will purchase pepsi three purchases from now ?

Method 1 :

$$P[\text{coke} \to ? \to ? \to \text{pepsi}] = P[\text{coke} \to \text{Pepsi} \to \text{Pepsi} \to \text{Pepsi}] + P[\text{coke} \to \text{Pepsi} \to \text{Coke} \to \text{Pepsi}] + P[\text{coke} \to \text{Coke} \to \text{Pepsi} \to \text{Pepsi}] + P[\text{coke} \to \text{Coke} \to \text{Coke} \to \text{Pepsi}]$$

$$= P[\text{pepsi}|\text{pepsi}] * P[\text{pepsi}|\text{pepsi}] * P[\text{pepsi}|\text{coke}] + P[\text{pepsi}|\text{coke}] * P[\text{coke}|\text{pepsi}] * P[\text{pepsi}|\text{coke}] + P[\text{pepsi}|\text{pepsi}] * P[\text{pepsi}|\text{coke}] * P[\text{coke}|\text{coke}] + P[\text{pepsi}|\text{coke}] * P[\text{coke}|\text{coke}] * P[\text{coke}|\text{coke}]$$

$$= 0.8 * 0.8 * 0.1 + 0.1 * 0.2 * 0.1 + 0.8 * 0.1 * 0.9 + 0.1 * 0.9 * 0.9$$

$$= 0.064 + 0.002 + 0.072 + 0.081$$

$$= 0.219.$$

Method 2 :

$$P^3 = P \times P \times P$$

$$= \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

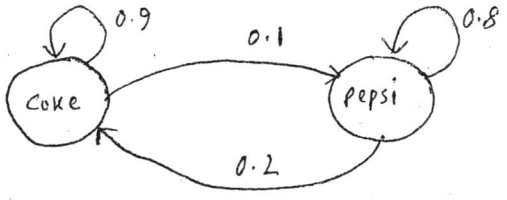
$$= \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix}$$

$$= \begin{matrix} \\ \text{coke} \\ \text{pepsi} \end{matrix} \begin{matrix} \begin{matrix} \text{coke} & \text{Pepsi} \end{matrix} \\ \begin{bmatrix} 0.781 & 0.219 \\ 0.438 & 0.562 \end{bmatrix} \end{matrix}$$

$$P[\text{coke} \to ? \to ? \to \text{pepsi}] = 0.219$$

Problem 3

Assume each person makes one cola purchase per week. Suppose 60% of all people now drink coke, and 40% drink pepsi. What fraction of people will be drinking coke three weeks from now?



Method 1:

$$P^3 = P \times P \times P$$

$$= \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

$$= \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix}$$

$$= \begin{matrix} \\ \text{Coke} \\ \text{Pepsi} \end{matrix} \begin{matrix} \begin{matrix} \text{Coke} & \text{Pepsi} \end{matrix} \\ \begin{bmatrix} 0.781 & 0.219 \\ 0.438 & 0.562 \end{bmatrix} \end{matrix}$$

$$P[\text{Coke} \to ? \to ? \to \text{Coke}] = 0.781$$

$$P[\text{Pepsi} \to ? \to ? \to \text{Coke}] = 0.438$$

Fraction of people = 60% of $P[\text{Coke} \to ? \to ? \to \text{Coke}]$

$+$ 40% of $P[\text{Pepsi} \to ? \to ? \to \text{Coke}]$

$$= 0.6 * 0.781 + 0.4 * 0.438$$

$$= 0.6438$$

Markov Models

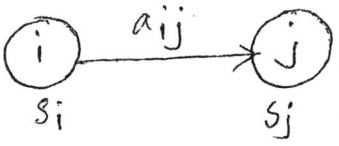
- set of states: $\{s_1, s_2, \cdots s_k, s_N\}$
- Markov chain property: probability of each subsequent state depends only on what was the previous state.

$$P(s_k \mid s_1 \cdots s_{k-1}) = P(s_k \mid s_{k-1})$$

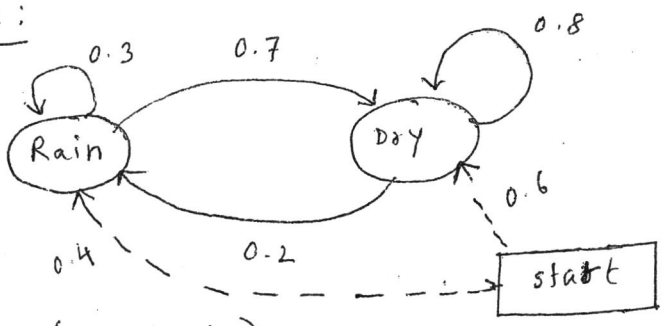
- Markov model:

Transition probabilities, $a_{ij} = P(s_j \mid s_i)$

Initial probabilities, $\pi_i = P(s_i)$



Example:



$P(Rain \mid Rain) = 0.3$

$P(Dry \mid Rain) = 0.7$

$P(Rain \mid Dry) = 0.2$

$P(Dry \mid Dry) = 0.8$

Find $P(Dry, Dry, Rain, Rain) = P(Rain \mid Rain) * P(Rain \mid Dry) * P(Dry \mid Dry) * P(Dry)$

$= 0.3 * 0.2 * 0.8 * 0.6$

$= 0.06 * 0.48$

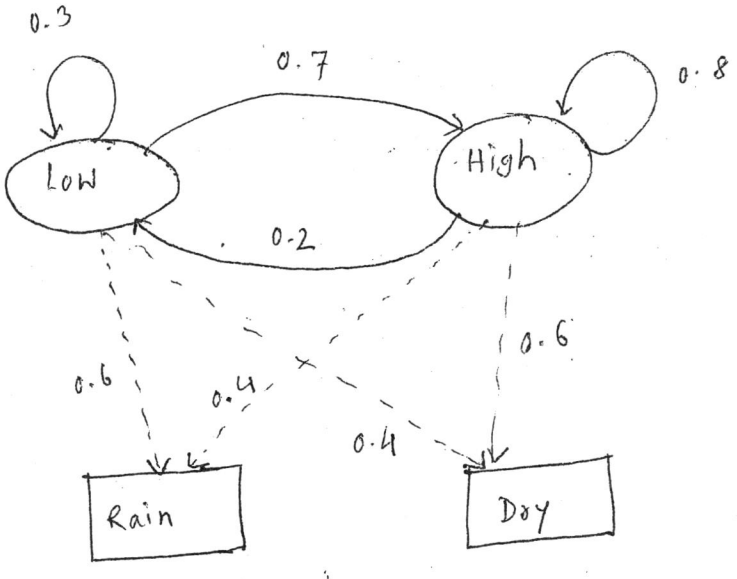
$= 0.0288$

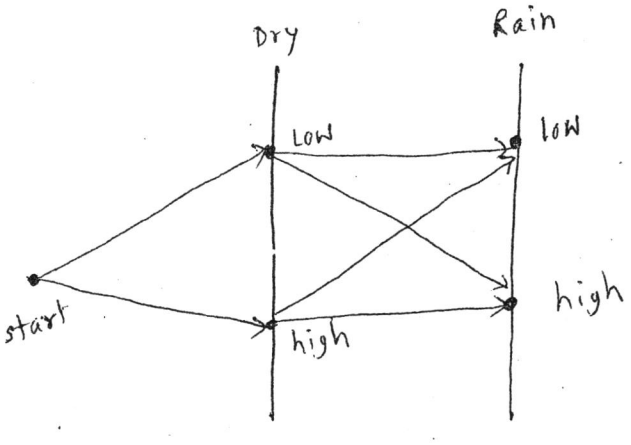
~~$P(S_k | S_1 S_2 \cdots S_{k-1})$ value:~~

$$P(S_1, S_2, S_3, \ldots, S_k) = P(S_k | S_1 S_2 \cdots S_{k-1}) P(S_1 S_2 \cdots S_{k-1})$$

$$= P(S_k | S_{k-1}) * P(S_{k-1} | S_{k-2}) P(S_1 S_2 \cdots S_{k-2})$$

$$= P(S_k | S_{k-1}) P(S_{k-1} | S_{k-2}) \cdots P(S_2 | S_1) P(S_1)$$





$$P(\text{Dry, Rain}) = P(\text{low}) * P(\text{Dry} | \text{low}) * P(\text{low} | \text{low}) * P(\text{Rain} | \text{low})$$
$$+ P(\text{low}) * P(\text{Dry} | \text{low}) * P(\text{high} | \text{low}) * P(\text{Rain} | \text{high})$$
$$+ P(\text{high}) * P(\text{Dry} | \text{high}) * P(\text{low} | \text{high}) * P(\text{Rain} | \text{low})$$
$$+ P(\text{high}) * P(\text{Dry} | \text{high}) * P(\text{high} | \text{high}) * P(\text{Rain} | \text{high})$$

$$= 2$$

Similarity Measure

	T1	T2	T3	T4	T5	T6	T7	T8
DOC1	0	4	0	0	0	2	1	3
DOC2	3	1	4	3	1	2	0	1
DOC3	3	0	0	0	3	0	3	0
DOC4	0	1	0	3	0	0	2	0
DOC5	2	2	2	3	1	4	0	2

Find similarity index using DOC2, DOC4 for

i) Cosine Similarity Measure

~~ii) Jaccard Similarity~~

~~iii) Pearson's correlation~~

Cosine Similarity Measure

$$\text{Dot_Product}(DOC2, DOC4) = \langle 3,1,4,3,1,2,0,1 \rangle * \langle 0,1,0,3,0,0,2,0 \rangle$$

$$= 3*0 + 1*1 + 4*0 + 3*3 + 1*0 + 2*0 + 0*2 + 1*0 = 10$$

$$\text{Norm}(DOC2) = \sqrt{3^2+1^2+4^2+3^2+1^2+2^2+0^2+1^2} = 6.4$$

$$\text{Norm}(DOC4) = \sqrt{0^2+1^2+0^2+3^2+0^2+0^2+2^2+0^2} = 3.74$$

$$\text{Cosine}(DOC2, DOC4) = \frac{\text{Dot_Product}(DOC2, DOC4)}{\text{Norm}(DOC2) * \text{Norm}(DOC4)}$$

$$= \frac{10}{6.4 * 3.74} = 0.42$$

So, DOC2 and DOC4 are 42% similar

Scanned with CamScanner

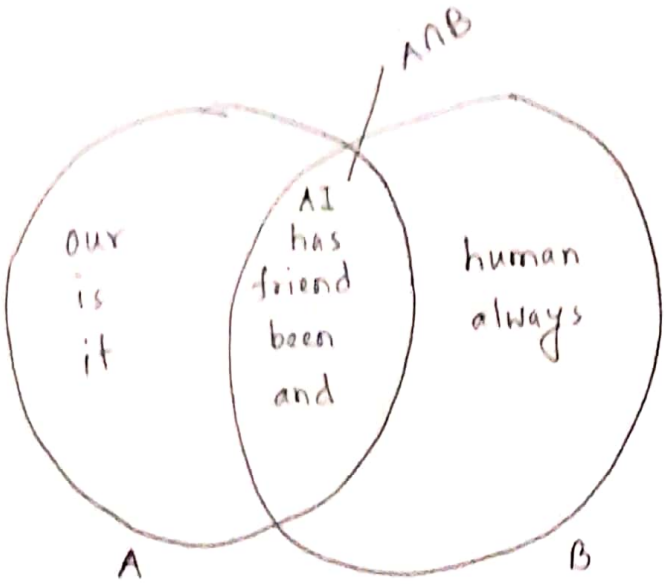
Jaccard Similarity

Ques: Sentence-I : AI is our friend and it has been friendly

Sentence-II : AI and humans have always been friendly

Find Jaccard similarity Index for sentence-I and sentence-II

Ans:



Lemmatization: (Reduce words to the same root word)

Here, friend, friendly → friend

has, have → has

$$J(\text{Sentence 1, Sentence 2}) = \frac{|A \cap B|}{|A \cup B|}$$

$$= \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

$$= \frac{5}{8+7-5}$$

$$= \frac{5}{10}$$

$$= 0.5$$

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...on's correlation

$$P(x,y) = \frac{S_{xy}}{S_x \cdot S_y}$$

$$= \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2} * \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}}$$

$$\bar{x} = \text{mean}(x) = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$\bar{y} = \text{mean}(y) = \frac{1}{n} \sum_{i=1}^{n} y_i$$

n is the number of attributes in x and y

Ques:

DOC2 = ⟨3, 1, 4, 3, 1, 2, 0, 1⟩

DOC4 = ⟨0, 1, 0, 3, 0, 0, 2, 0⟩

Find pearson correlation index

Ans:

$x = \langle 3, 1, 4, 3, 1, 2, 0, 1 \rangle$

$y = \langle 0, 1, 0, 3, 0, 0, 2, 0 \rangle$

$$\therefore \bar{x} = \frac{3+1+4+3+1+2+0+1}{8} = \frac{15}{8} = 1.875$$

$$\therefore \bar{y} = \frac{0+1+0+3+0+0+2+0}{8} = 0.75$$

$$\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = (3-1.875) * (0-0.75) + (1-1.875) * (1-0.75) + (4-1.875) * (0-0.75) + (3-1.875) * (3-0.75) + (1-1.875) * (0-0.75) + (2-1.875) * (0-0.75) + (0-1.875) * (2-0.75) +$$

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$= (1-1.875) \times (0-0.75)$

$= ?$

$$\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2} = \sqrt{(3-1.875)^2+(1-1.875)^2+(4-1.875)^2+(3-1.875)^2+(1-1.875)^2+(2-1.875)^2+(0-1.875)^2+(1-1.875)^2}$$

$= ?$

$$\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2} = \sqrt{(0-0.75)^2+(1-0.75)^2+(0-0.75)^2+(3-0.75)^2+(0-0.75)^2+(0-0.75)^2+(2-0.75)^2+(0-0.75)^2}$$

$= ?$

$$\therefore p(x,y) = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}$$

$= ?$

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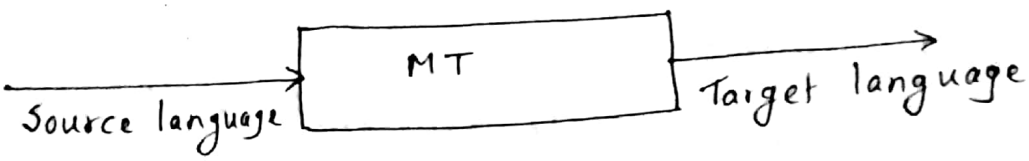
Dr. Mohammad Nurul Huda
Statistical Machine Trans
Lecture sheet - I

Machine Translation (MT):

MT can be defined as the use of computers to automate some or all of the process of translating from one language to another.



MT uses ideas and techniques

- Linguistics
- Computer Science
- Artificial Intelligence (AI)
- Translation theory
- Statistics

Commercially Interesting:

- US has invested in MT for intelligence purposes.
- EU spends more than €1,000,000,000 on translation costs each year.
- MT is used in UNL (Universal Network Language)
- MT is used to translate technical documents, reports, instruction manuals etc.

Objective:

- The goal is not fluent translation
- Only correct and understandable output.

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Various approaches to Machine Translation:

- The transfer approach
- The Interlingua approach
- The direct approach
- Corpus-based approach
- Word-for-word translation
- statistical translation

Word-for-word translation:

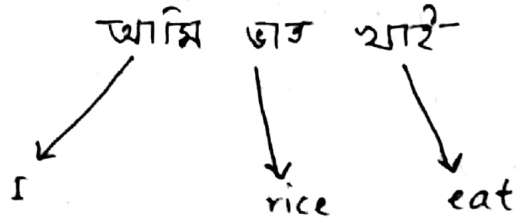
- Use a machine-readable bilingual dictionary to translate each word in a text.

Dictionary:

Bangla	English
আমি	I
ভাত	rice
খাই-	eat
.	.
.	.
.	.

Bangla: আমি ভাত খাই-

English: ?



English: I rice eat

Advantage:

Easy to implement, results give a rough idea about what the text is about.

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vantage:

Problems with word order means that this results in low-quality translation.

The Transfer Approach:

The transfer model involves three stages

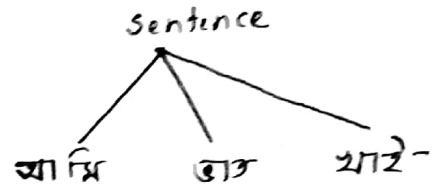
- Analysis
- transfer
- generation

Analysis:

The source language sentence is parsed, and the sentence structure and the constituents of the sentence are identified.

Example:

আমি ভাত খাই-

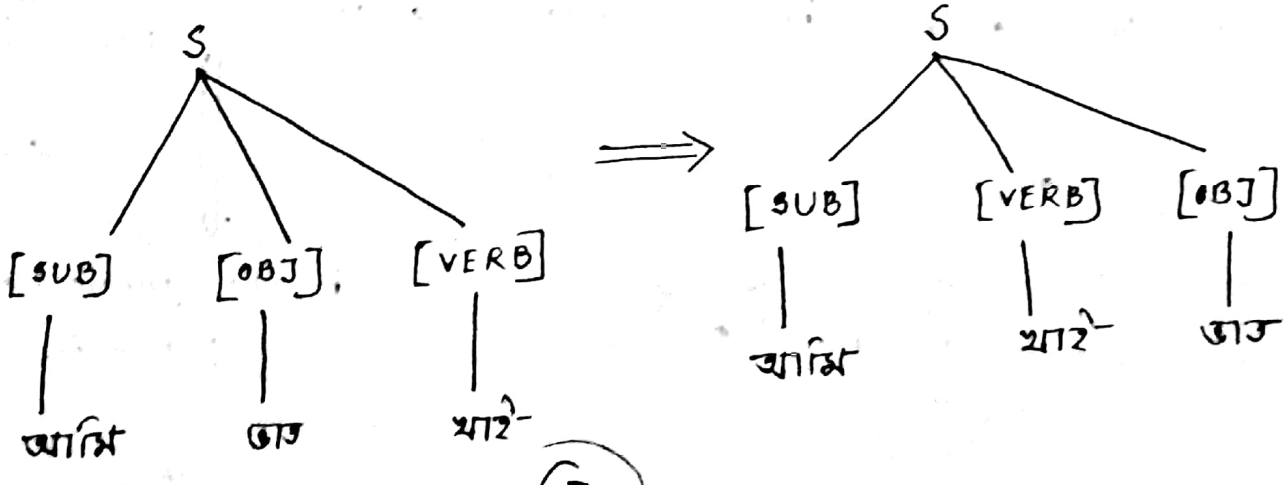


Words: আমি, ভাত, খাই-

Sentence structure: [subject] [object] [verb]

Transfer stage:

Transformations are applied to the source language parse tree to convert the structure to that of the target language.



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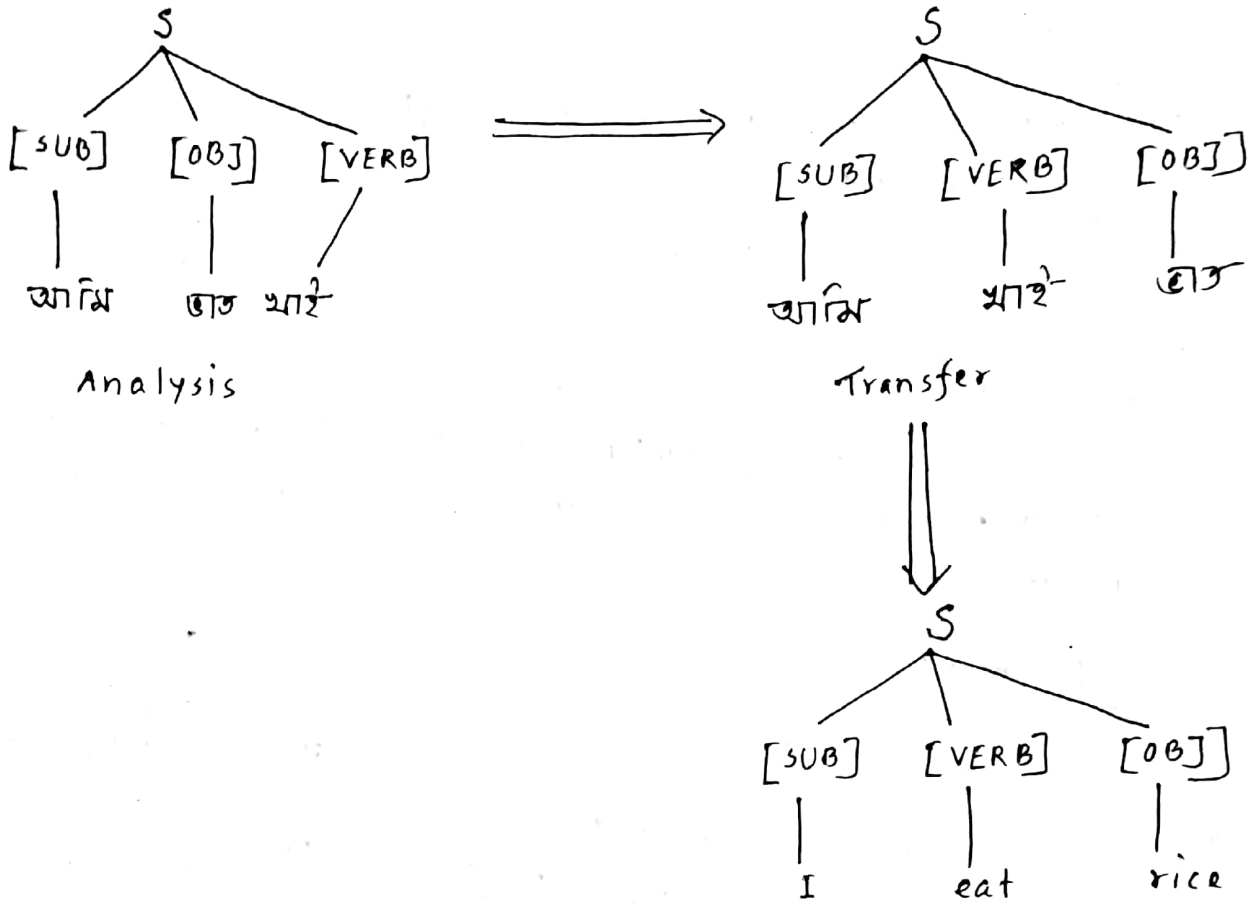
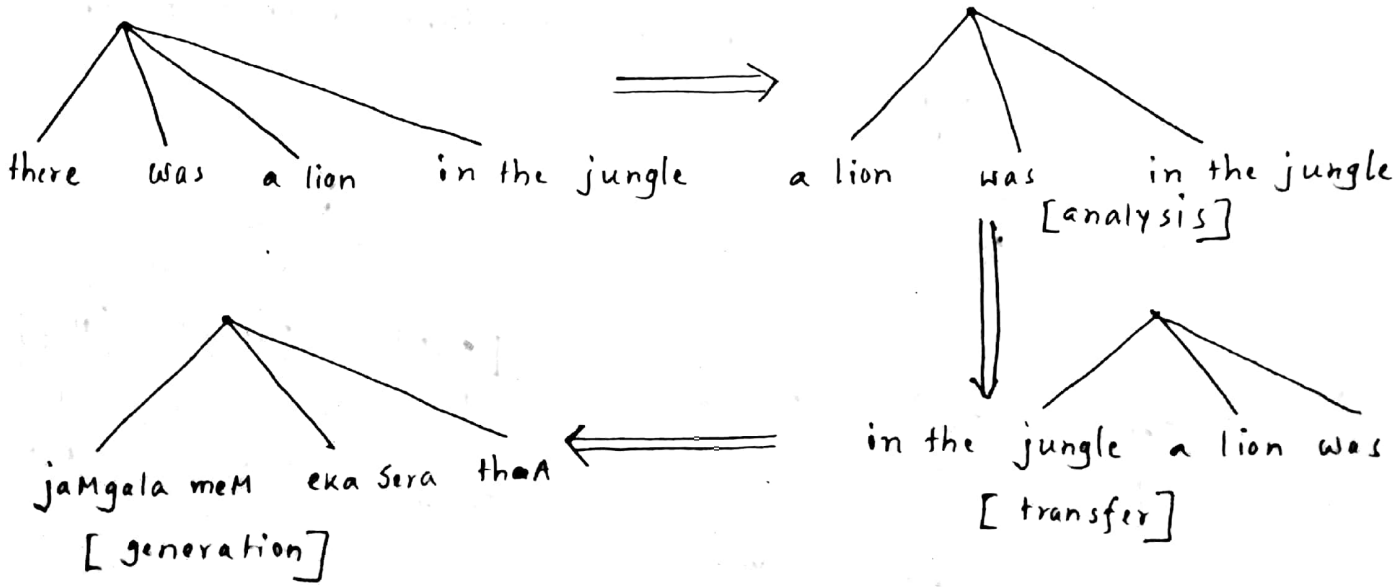
The generation stage :

Translates the words and expresses the tense, number, gena etc. in the taget language.



Example :

S : There was a lion in the jungle



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Interlingua Approach:

It is a two stage process.

1. extracting the meaning of a source language sentence in a language-independent form.
2. Generating a target language sentence from the meaning.

Importance:

It expresses the complete meaning of any sentence by using a set of universal concepts and relations.

Example:

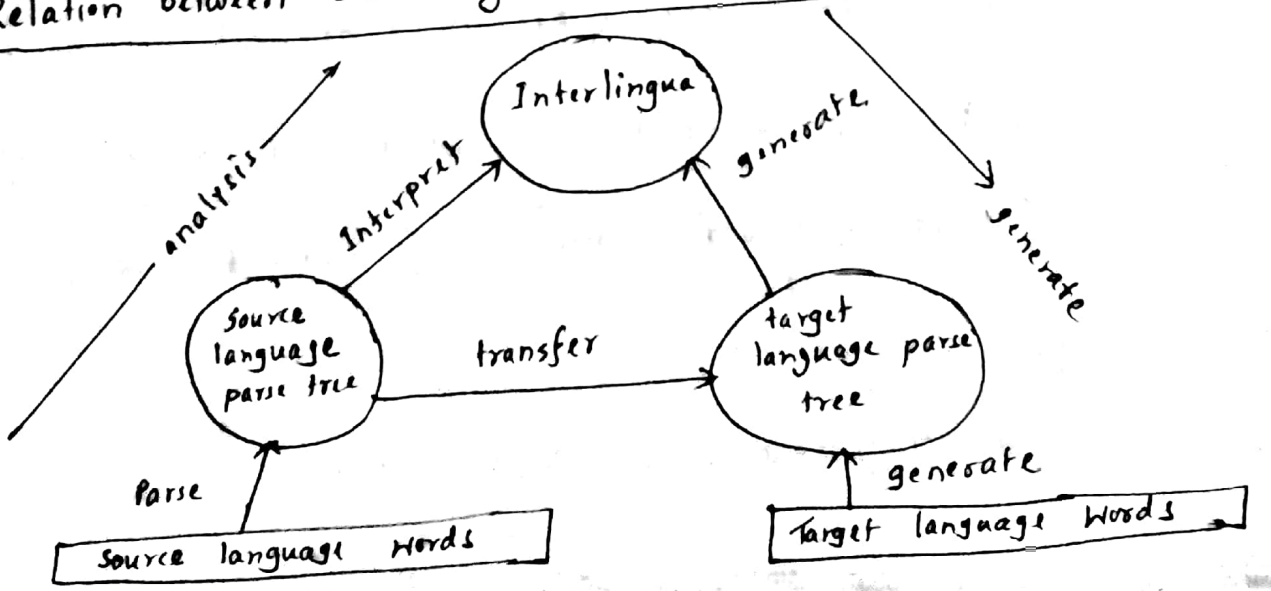
The child broke the toys with a hammer.

Interlingual Representation

PREDICATE	BREAK
AGENT	CHILD number: SING
THEME	TOY number: PLUR
INSTRUMENT	HAMMER number: SING
TENSE	PAST

⟨AGENT⟩ ⟨⟨PREDICATE⟩ ⟨TENSE⟩⟩ ⟨THEME⟩ ⟨INSTRUMENT⟩

Relation between Interlingua and Transfer approach:



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The Direct Approach:

step1: Morphological analysis

step2: Identify constituents

step3: Reorder them according to the constituent order in target language

step3: look up the words in an source-target language dictionary

step4: Inflect the target language words appropriately.

Example:

We will demand peace in the country.

English Sentence:

We will demand peace in the country.

Morphological Analysis:

We demand FUTURE peace in country

Identify constituents:

<We> <demand FUTURE> <peace> <in country>

Reorder according to target language:

<We> <in country> <peace> <demand FUTURE>

Look-up in the source-target language dictionary

<আমরা> <দেশের মধ্যে> <শান্তি> <দাবি FUTURE>

Inflect:

আমরা দেশের শান্তি দাবি করব।

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pus-Based Approaches:

- Not use human-encoded linguistic knowledge to solve the translation problem
- Used a training corpus of already translated texts - a parallel corpus - to guide the translation process.

A Parallel Corpus

Source language collection	Target language collection

Example Based Machine Translation:

- Split the problem into subproblems.
- Recall how they solved similar subproblems in the past.
- Adapt these solutions to the new situation
- Combine the solutions to solve the bigger problem.

Example:

He bought a huge house.

Corpus:

1a) He bought a pen

1b) সে একটি কলম কিনেছিল

2a) All ministers have a huge house

2b) সব মন্ত্রীর একটি বড় বাড়ি আছে

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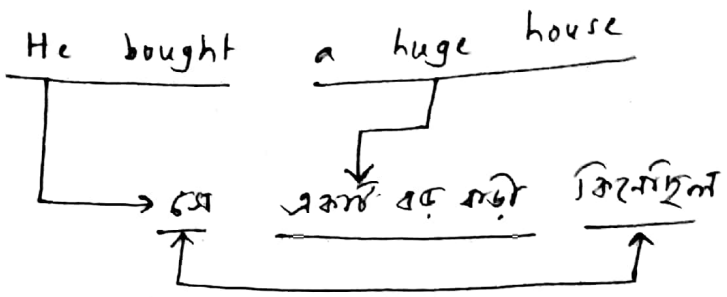


Example based MT (EBMT) entails three steps:

1. Matching fragments against the parallel corpus.
2. Adapting the matched fragments to the target languege.
3. Recombing these translated fragments appropriately.

Why machine Translation is hard?:

- Word order
- Word sense
- Pronouns
- Tense
- Idioms

Word order:

English: SUB- VERB- OBJECT

বাংলা: SUB- OBJECT- VERB

English: I eat rice

বাংলা:

Sense ambiguity:

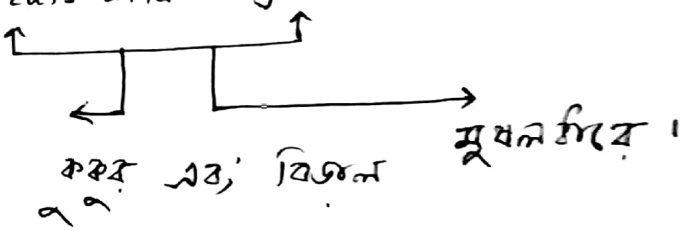
Example 1:
- Bank as in river
- Bank as in financial Institution

Example 2:
- Plant as in a tree
- Plant as in a factory

** Different word senses will likely translate into different words in another language.

Idioms:

It has been raining cats and dogs



Which one will be appropriate
- বিড়াল এবং কুকুর
- মুষলধারে।

Pronouns:

When should we use 'she' or 'he'.

সে ভাত খায় — He eats rice
সে ভাত খায় — She eats rice
} Which one is correct?

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Resources for Interlingual machine translation system

- Dictionaries (For analysis and generation)
- A conceptual knowledge based lexicon
- A set of projection rules
- Grammars for the analysis and generation of the languages involved

Characteristics of Interlinguas

- Unambiguous
  - should be more explicit.
- Represent the full content of the input text
  - Morphological
  - Syntactic
  - Semantic
  - Pragmatic char c
- should be Universal
  - Abstract meaning of any text.
- should represent the content of the input alone
- should be independent of both the source language text and the target language text.

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Interlingua:

Interlingua machine translation is a classical approach.

Two steps of Interlingua:

1. Source language text is transformed into an interlingua (Here, interlingua is an abstract language-independent representation)
2. Target language is generated from the Interlingua.

[Alternative machine translation approach to the direct approach and the transfer approach]

Advantages of Interlingual approach:

- Requires fewer components in order to relate each source language to each target language.
- Takes fewer components to add a new language.
- Supports paraphrases of the input in the original language
- Allows both the analysers and generatoos to be written by monolingual system developers.
- Handles languages that are very different from each other

Disadvantages of Interlingual approach:

- Definition of an interlingua is difficult.

Objective of Interlingua approach:

Analysis of text of a source language with respect to

i) Morphology
ii) Syntax
iii) Semantic

} Meaning

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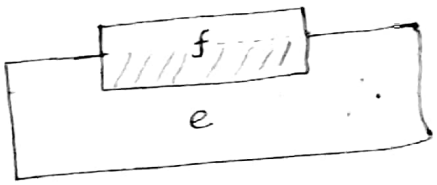
that the best translation is the sentence that is both faithful to the original sentence and fluent in the target language

Ans:

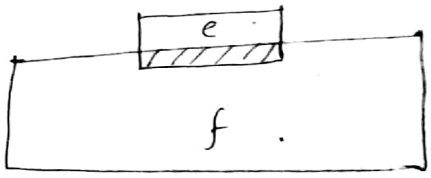
$$P(f|e) = \frac{\text{Total } f \text{ in } e}{\text{Total } e}$$



$$= \frac{P(f \wedge e)}{P(e)}$$

$$\Rightarrow P(f \wedge e) = P(f|e)\, P(e) \quad \cdots \quad eq^{(1)}$$

$$P(e|f) = \frac{\text{Total } e \text{ in } f}{\text{Total } f}$$



$$= \frac{P(f \wedge e)}{P(f)}$$

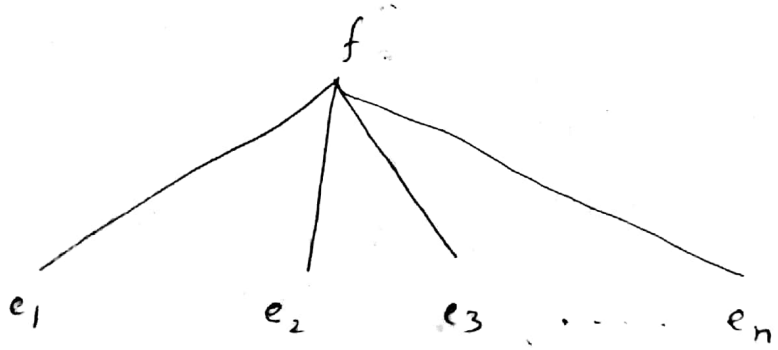
$$\Rightarrow P(f \wedge e) = P(e|f)\, P(f) \quad \cdots \quad eq^{(2)}$$

From $eq^{(1)}$ and $eq^{(2)} \Rightarrow$

$$P(e|f)\, P(f) = P(f|e)\, P(e)$$

$$\Rightarrow P(e|f) = \frac{P(f|e)\, P(e)}{P(f)}$$

If there are n sentences in target language for source language f



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$$P(e_1|f) = \frac{P(f|e_1)\ P(e_1)}{P(f)}$$

$$P(e_2|f) = \frac{P(f|e_2)\ P(e_2)}{P(f)}$$

$$P(e_3|f) = \frac{P(f|e_3)\ P(e_3)}{P(f)}$$

$$P(e_n|f) = \frac{P(f|e_n)\ P(e_n)}{P(f)}$$

$$P(e|f) = \max\left\{ P(e_1|f),\ P(e_2|f),\ P(e_3|f),\ \ldots\ldots,\ P(e_n|f) \right\}$$

$$= \max\left\{ \frac{P(f|e_1)\,P(e_1)}{P(f)},\ \frac{P(f|e_2)\,P(e_2)}{P(f)},\ \ldots\ldots,\ \frac{P(f|e_n)\,P(e_n)}{P(f)} \right\}$$

$$\cong \max\left\{ P(f|e_1)\,P(e_1),\ P(f|e_2)\,P(e_2),\ \ldots\ldots,\ P(f|e_n)\,P(e_n) \right\}$$

$$= \max\left\{ P(f|e_i)\ P(e_i) \right\}$$

where $i = 1 \ldots n$

$$\hat{e} \in \{e_1, e_2, \ldots e_n\}$$

$$\hat{e} = \arg\max_{e_i} \left\{ P(f|e_i)\ P(e_i) \right\}$$

- $P(f|e_i)$ → Faithful sentence for source text
- $P(e_i)$ → fluent sentence in target language

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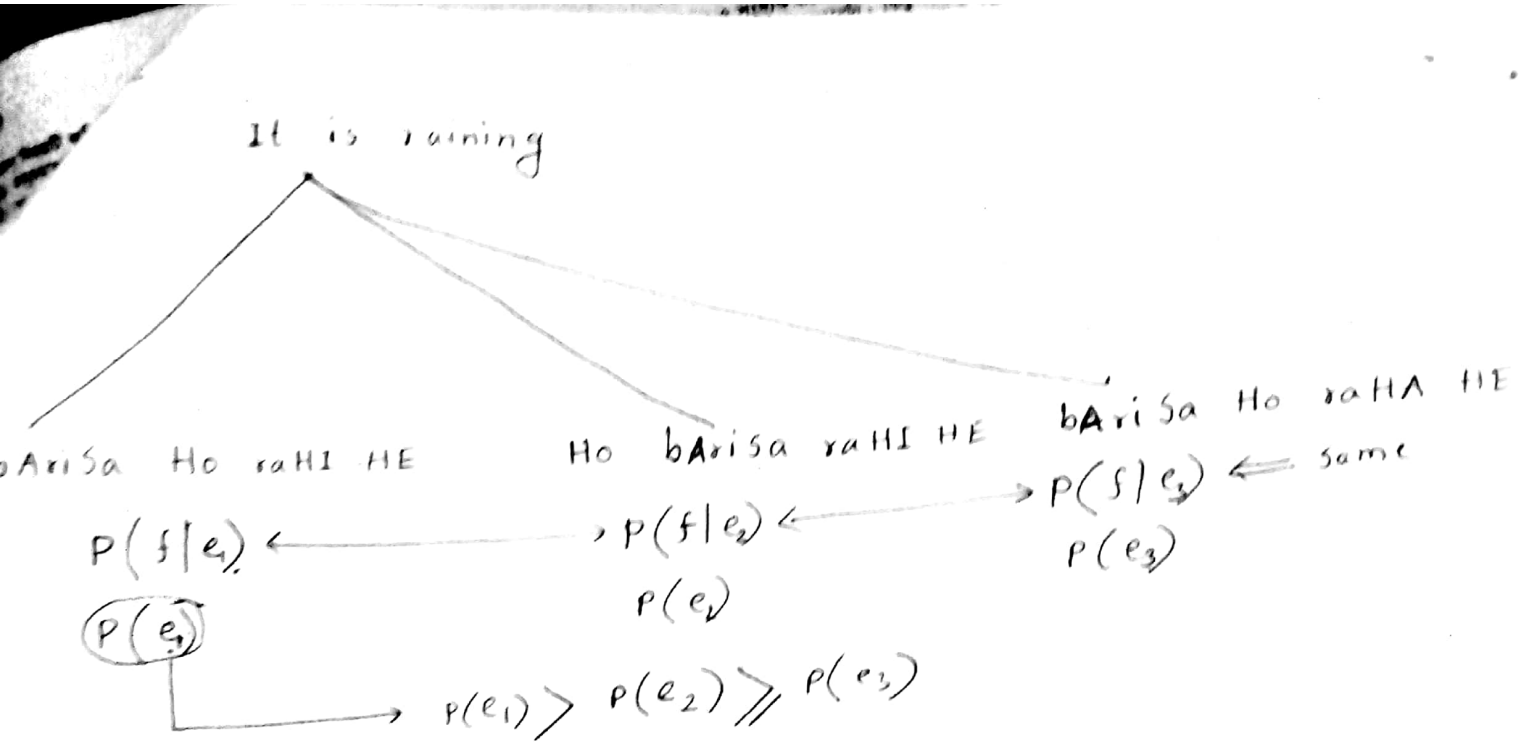

Since $P(f|e_1)\,P(e_1)$ maximum

It is raining $\Longrightarrow$ bArisa Ho raHI HE

Noisy channel Model for Machine Translation :

Let source sentence f to be translated was initially conceived in language E as some target sentences e.

During communication e was corrupted by the channel to f

Each sentence in E is a translation of f with some probability

The sentence we choose as the translation ($\hat{e}$) is the one that has the highest probability

$$\hat{e} = \underset{e}{\operatorname{argmax}}\ P(e|f)$$

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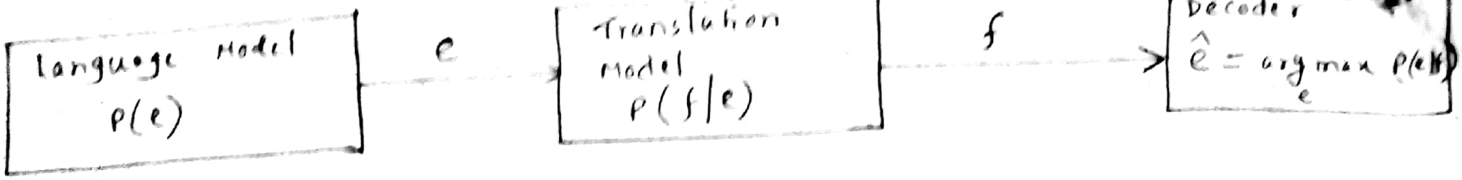


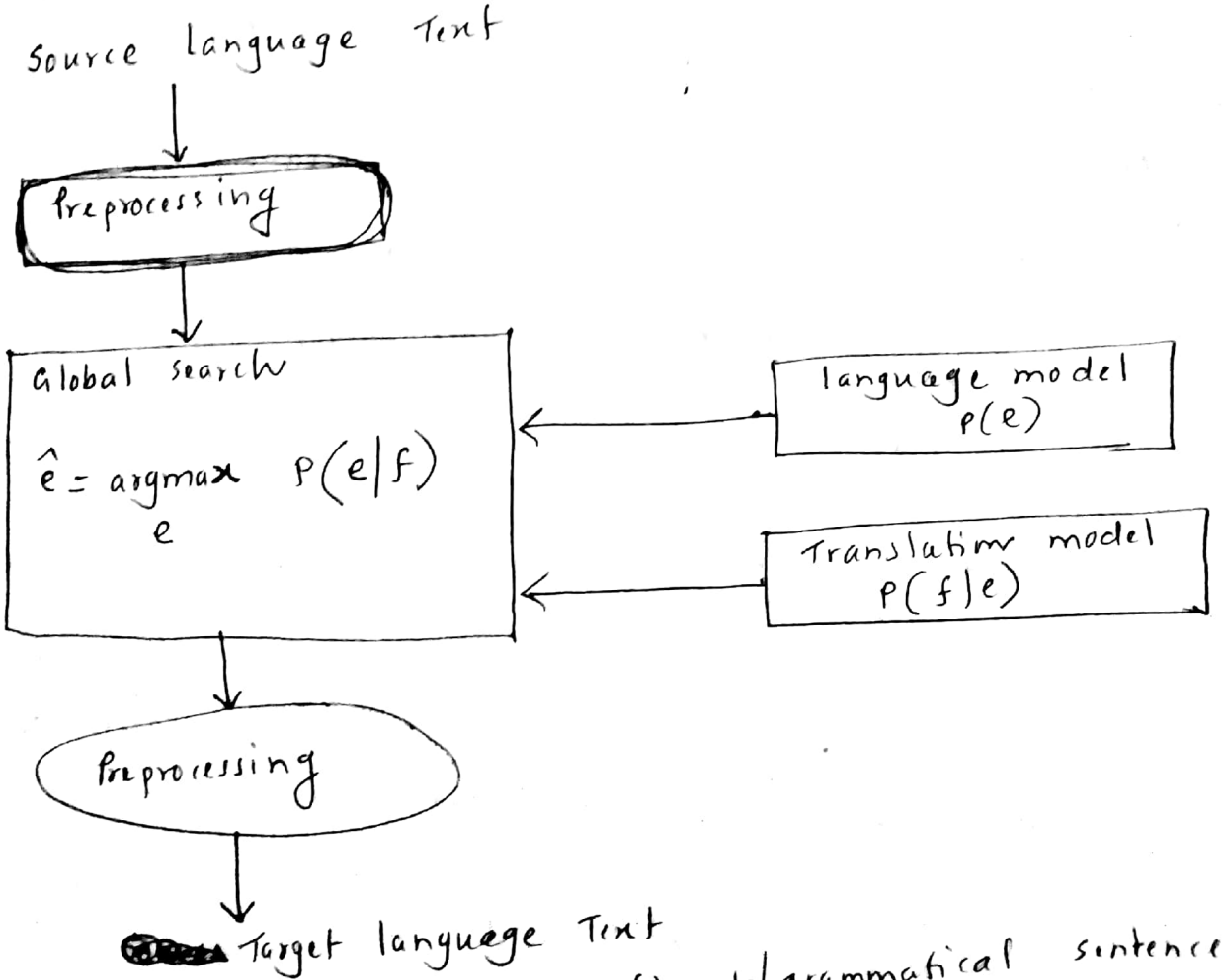
P(e): language model

The kind of sentences that are likely in the language E.

P(f|e): Translation model

The way sentences in E get converted to sentences in F. This is called translation model.

System diagram of Machine Translation:



P(e): Assigns higher probability to fluent/grammatical sentences
: Estimated using monolingual corpora

P(f|e): Assign higher probability to sentences that have corresponding meaning
: Estimated using bilingual corpora

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Lexical Translation

Haus – house, building, home, household, shell

German Text — English Text

Step1: Collect statistics

Look at a parallel corpus (German Text vs English translation)

Haus	Count
house	8000
building	1600
home	200
household	150
shell	50

Step2: Estimate translation probabilities

Total = 8000 + 1600 + 200 + 150 + 50

= 10000

$$P_{Haus}(house) = \frac{8000}{10000} = 0.8$$

$$P_{Haus}(building) = \frac{1600}{10000} = 0.16$$

$$P_{Haus}(home) = \frac{200}{10000} = 0.02$$

$$P_{Haus}(household) = \frac{150}{10000} = 0.015$$

$$P_{Haus}(shell) = \frac{50}{10000} = 0.005$$

$$P_f(e) = \begin{cases} 0.8 & \text{if } e = \text{house} \\ 0.16 & \text{if } e = \text{building} \\ 0.02 & \text{if } e = \text{home} \\ 0.015 & \text{if } e = \text{household} \\ 0.005 & \text{if } e = \text{shell} \end{cases}$$

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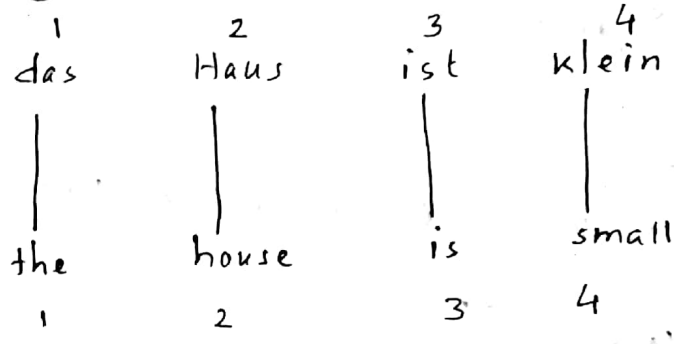
Alignment:

In a parallel text (or when we translate), we align words in one language with the words in the other.



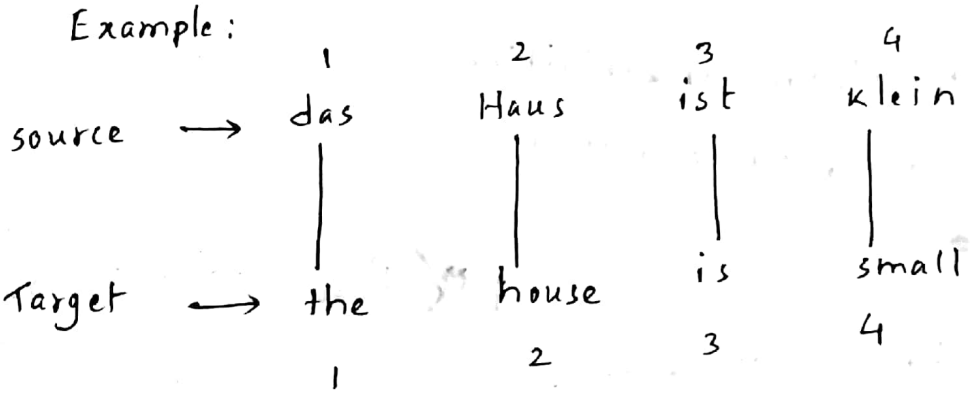
Word positions are numbered 1–4

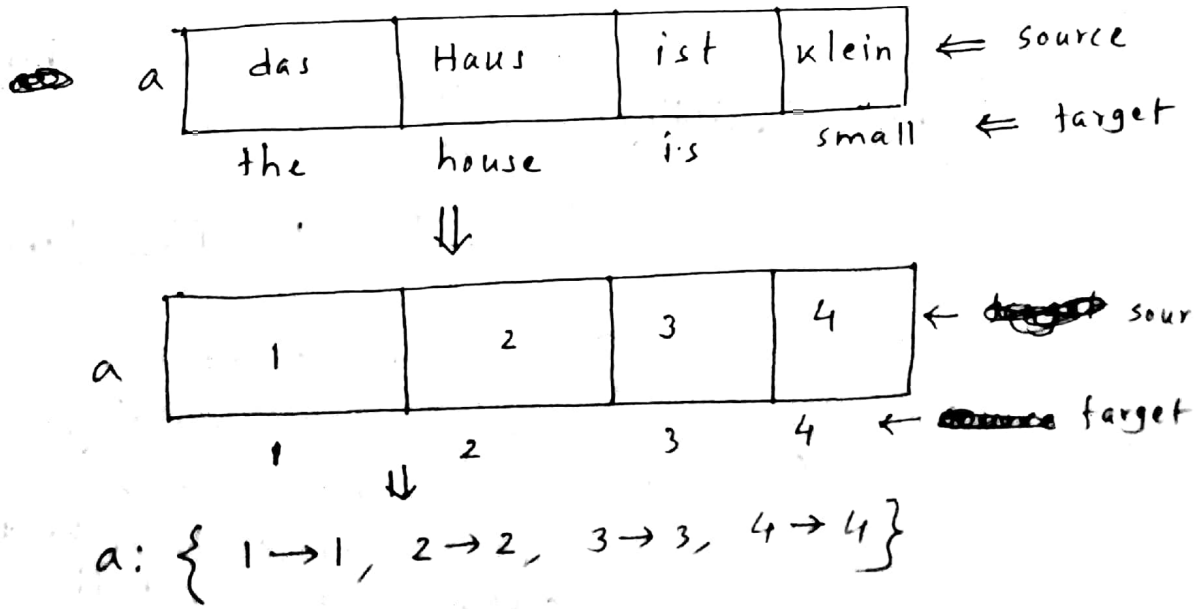
Alignment function

Mapping an target word (English) at position i to a source word (German) at position j with a function

a : i → j

Example:





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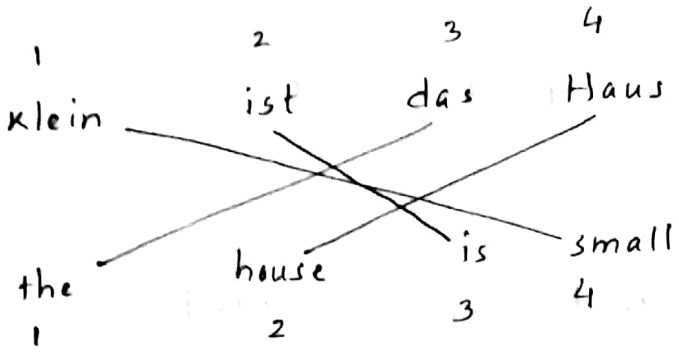
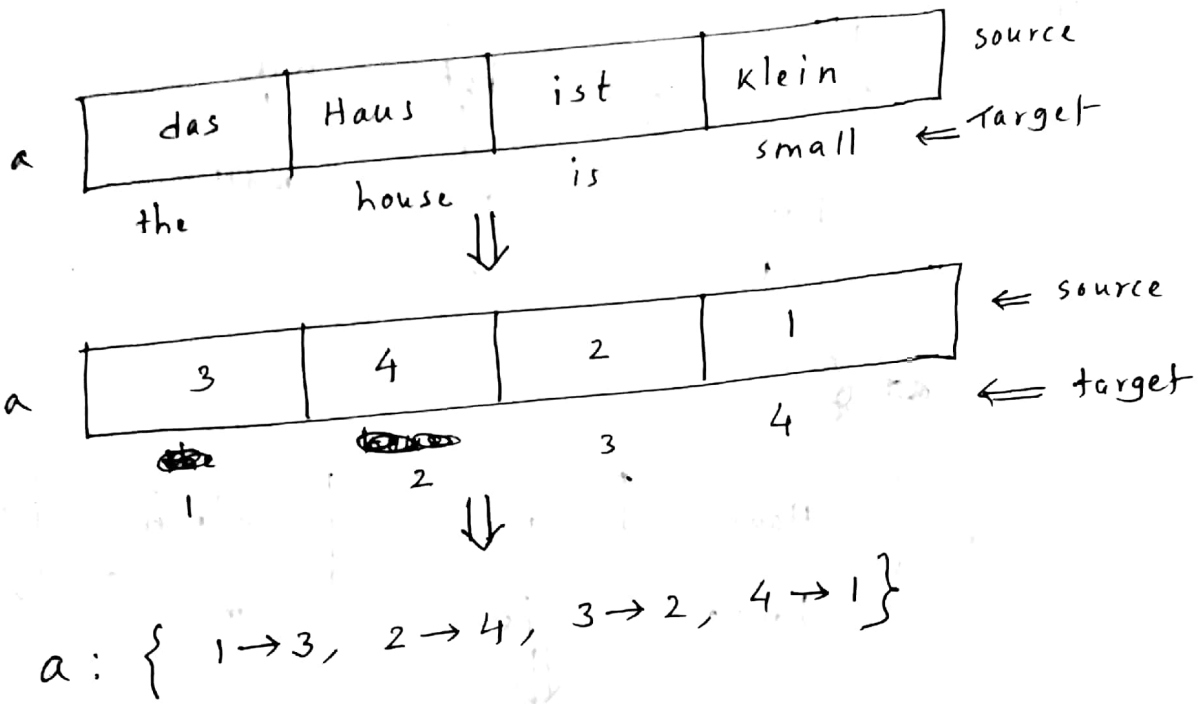
ordering:

Words may be reordered during translation

Example:



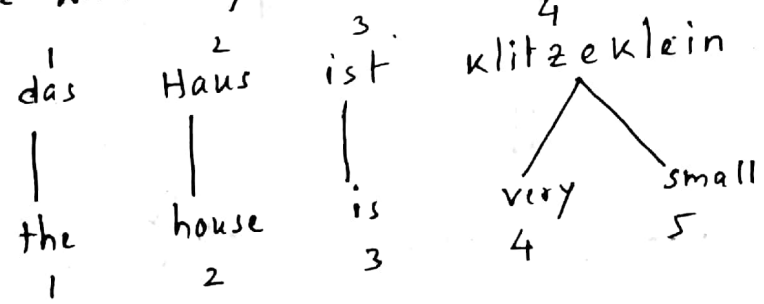


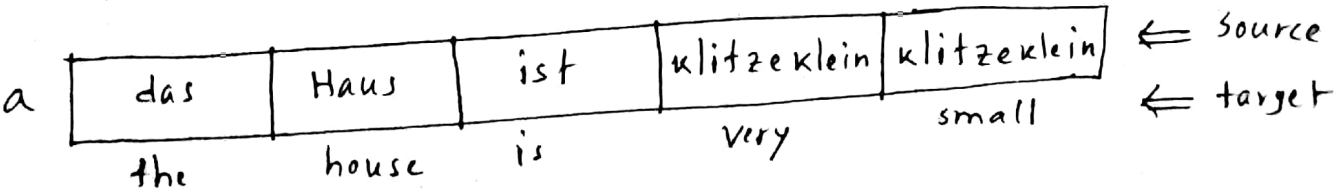
$a: \{ 1 \to 3,\ 2 \to 4,\ 3 \to 2,\ 4 \to 1 \}$

One-to-many translation:

A source word may translate into multiple target words

Example:





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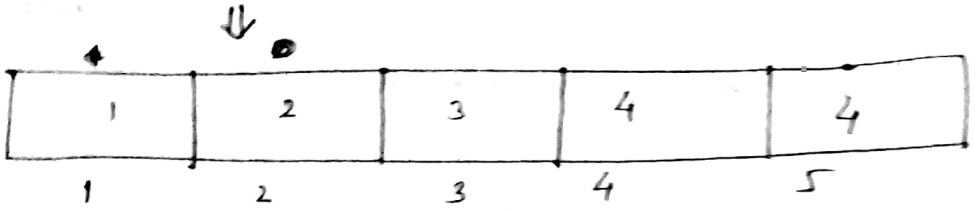
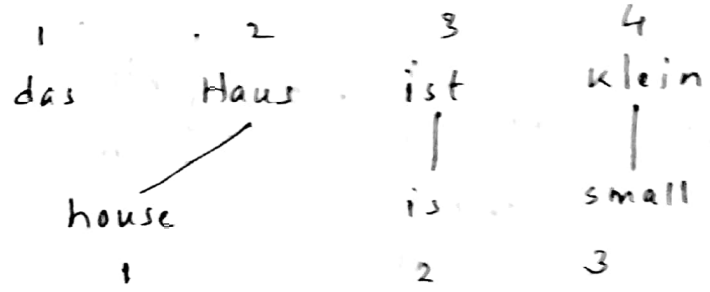


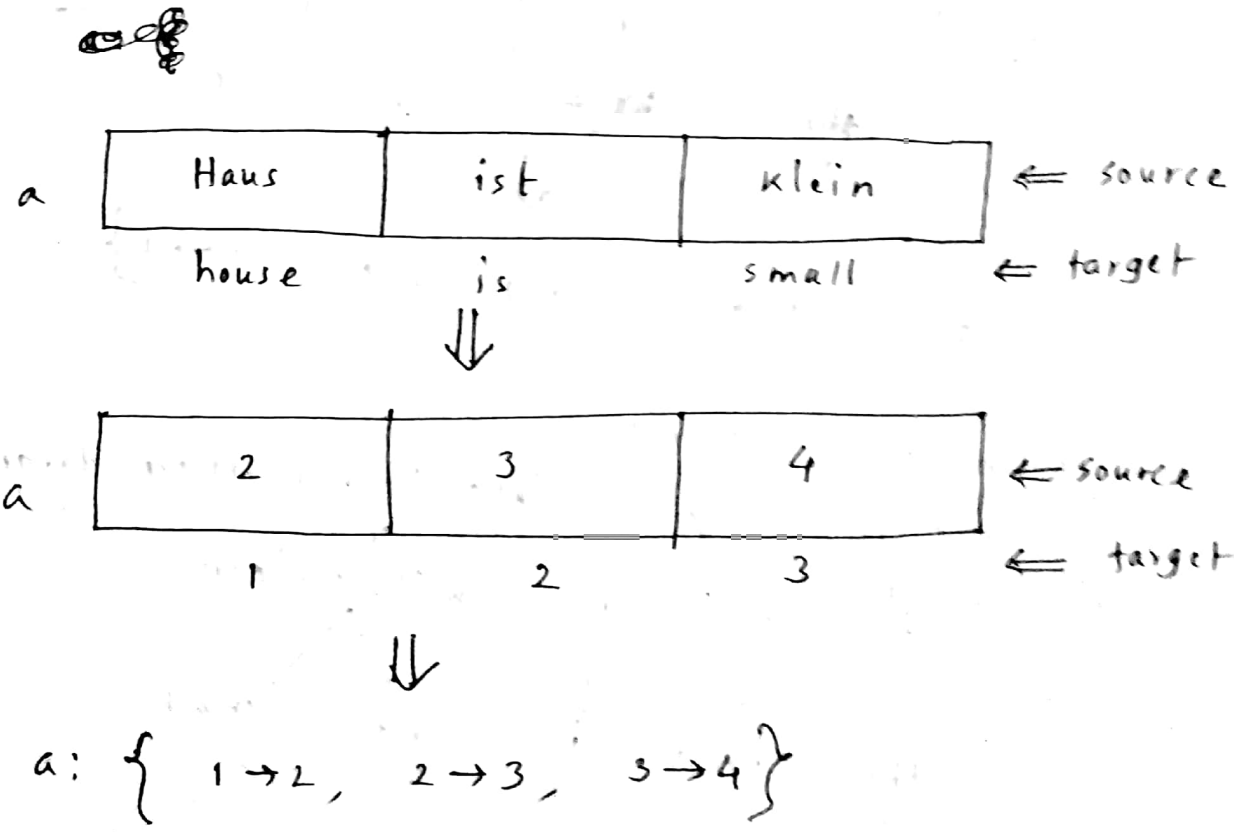
a : { 1→1, 2→2, 3→3, 4→4, 5→4 }

Dropping Words :

Words may be dropped when translated

Example :





a : { 1→2, 2→3, 3→4 }

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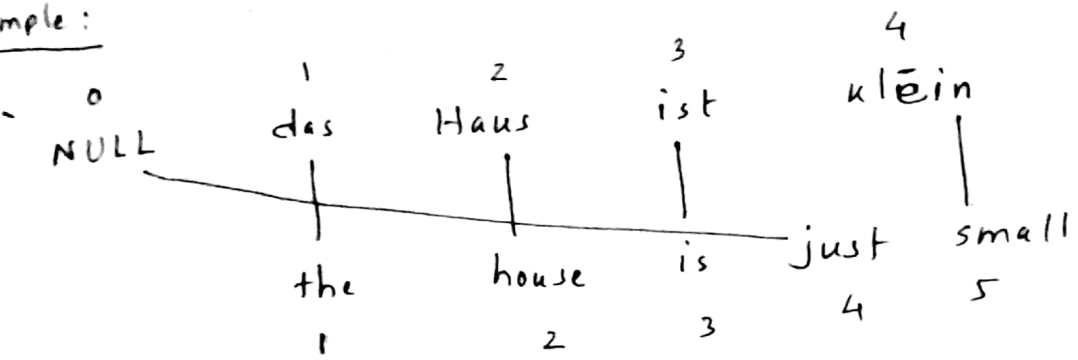
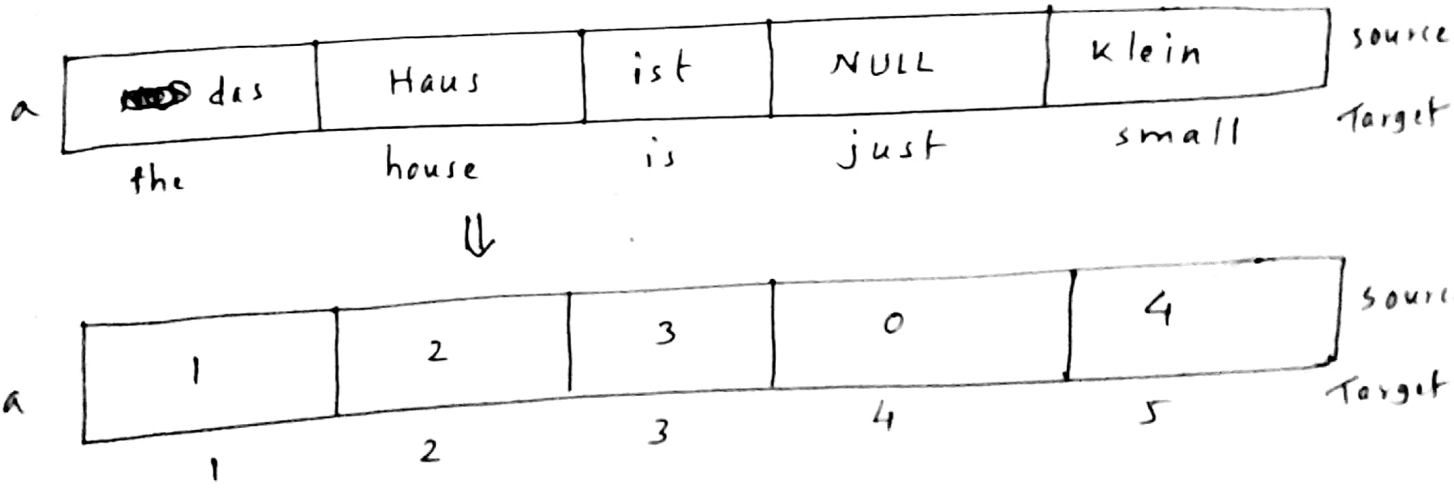
Inserting Words:

Words may be added during translation

- The English just does not have an equivalent in German
- We still need to map it to something: special NULL token

Example:





$a: \{1 \to 1,\ 2 \to 2,\ 3 \to 3,\ 4 \to 0,\ 5 \to 4\}$

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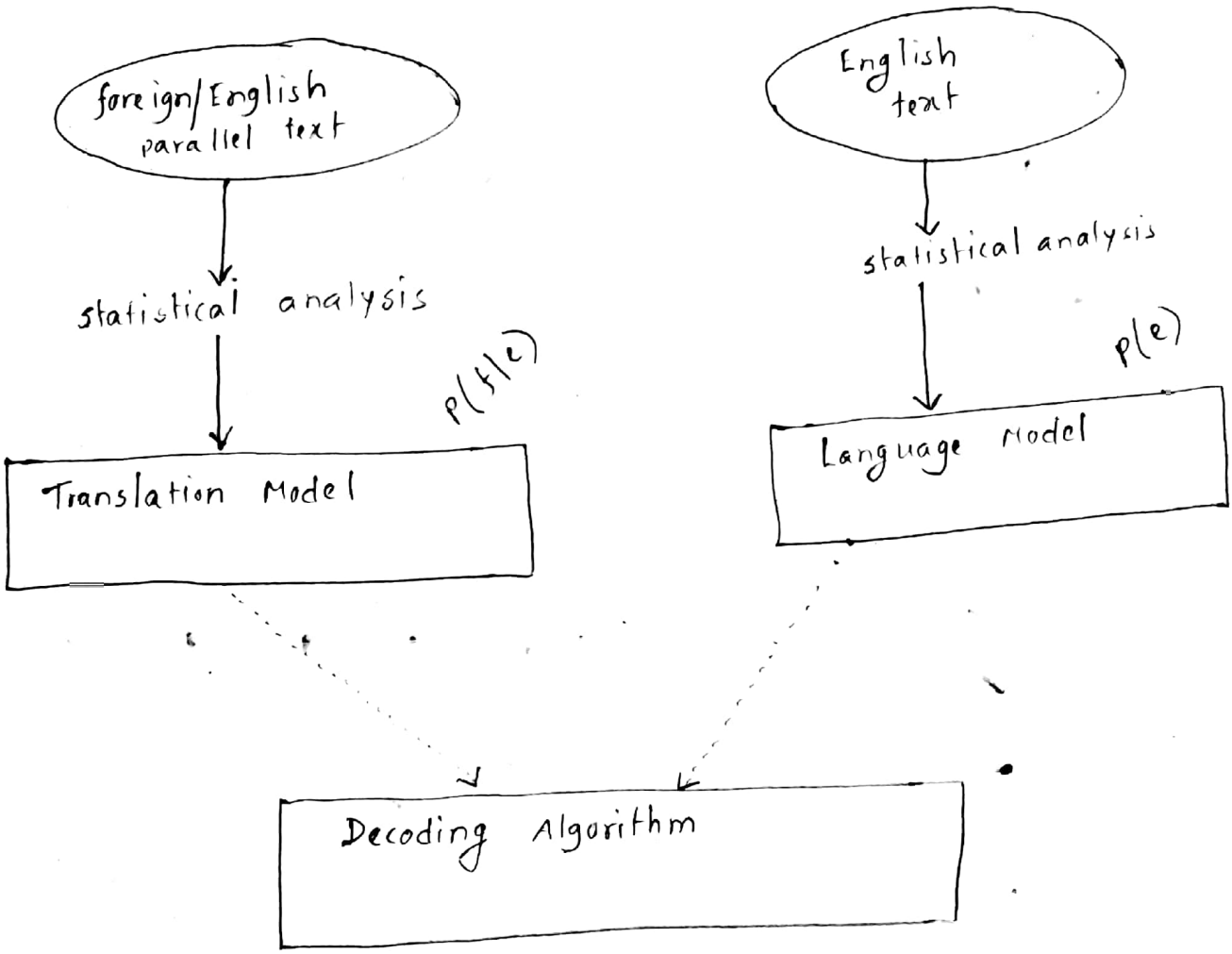
Statistical Machine Translation:

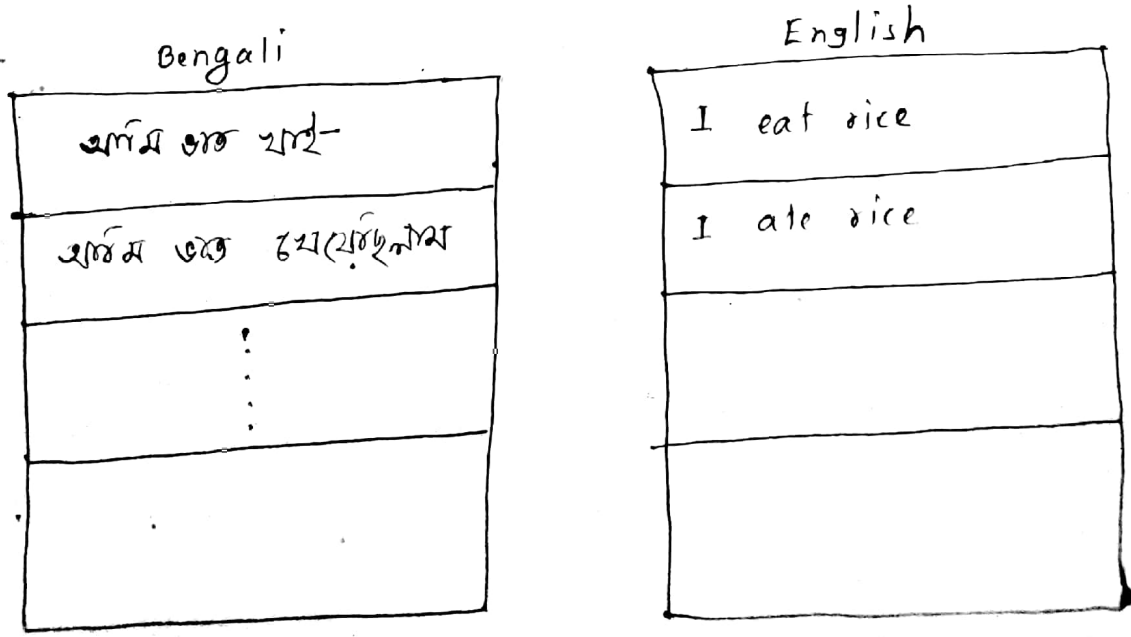
$$P(e|f) = \max\left(P(f|e) * P(e) \right)$$

$P(e)$ → Language model

$P(f|e)$ → Translation model



Parallel Corpus:



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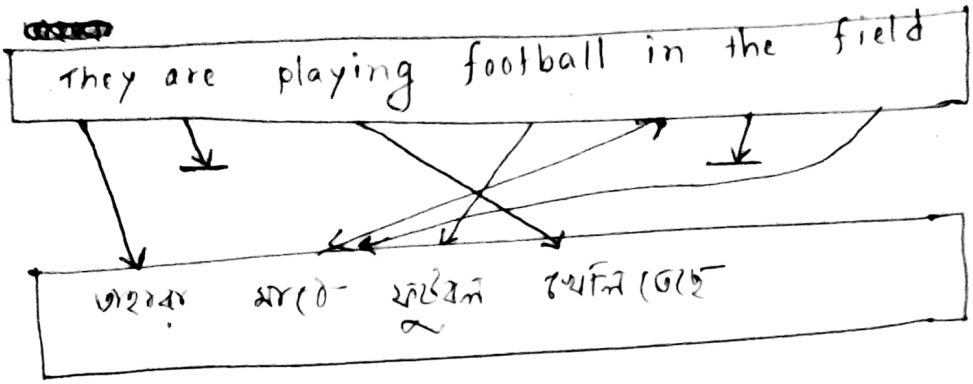
Different Models in statistical Machine Translation

- Word based models.
- Phrase based models.
- Sytax based models.

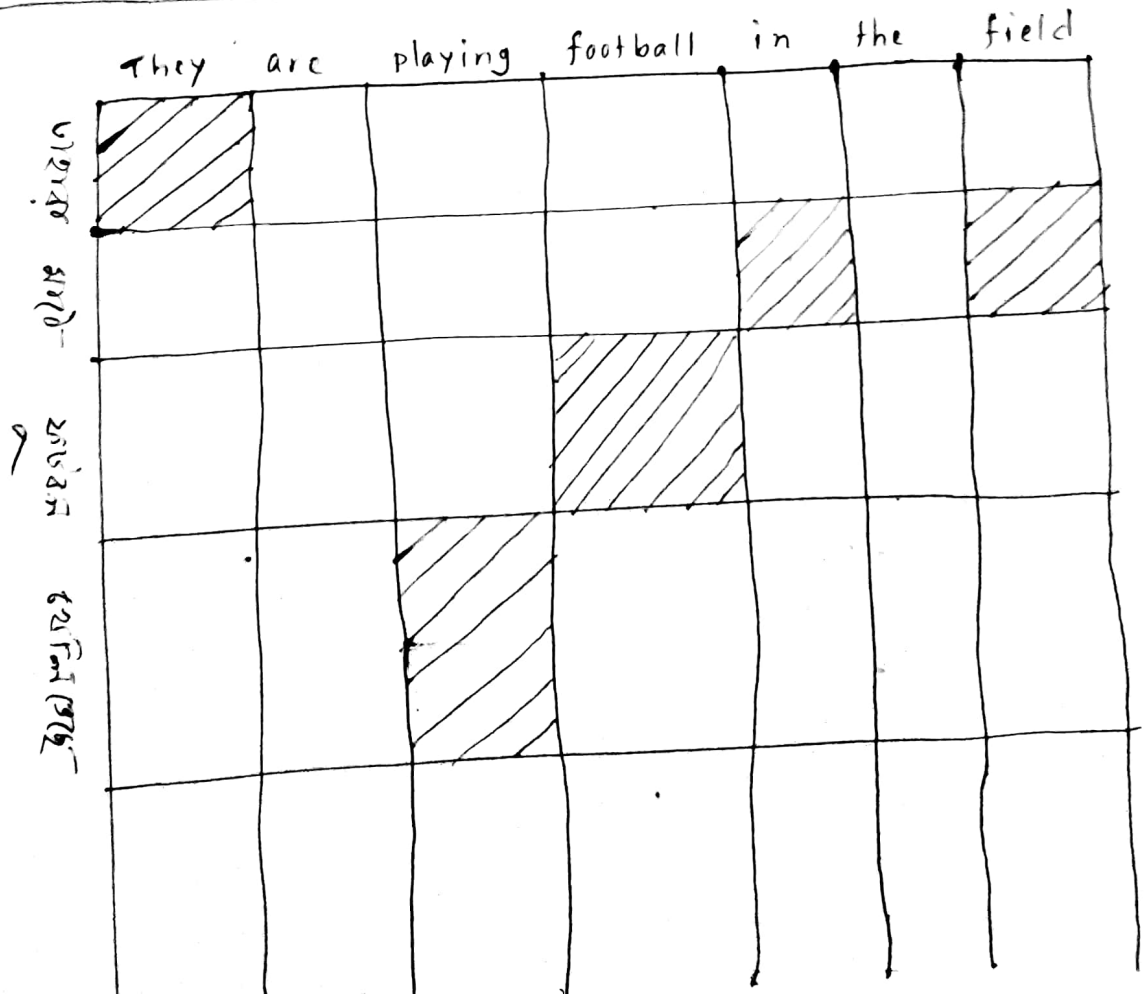
Word based models:

Translation process is decomposed into smaller steps, each is tied words.

Example:



Word alignment:



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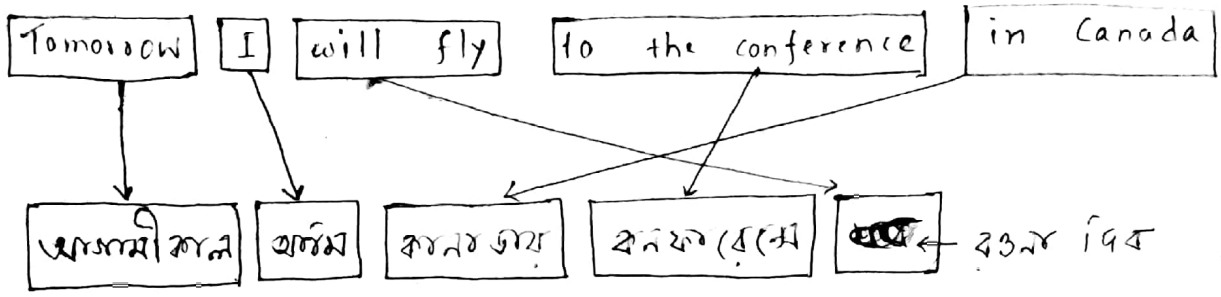
phrase-based models:

Ex:



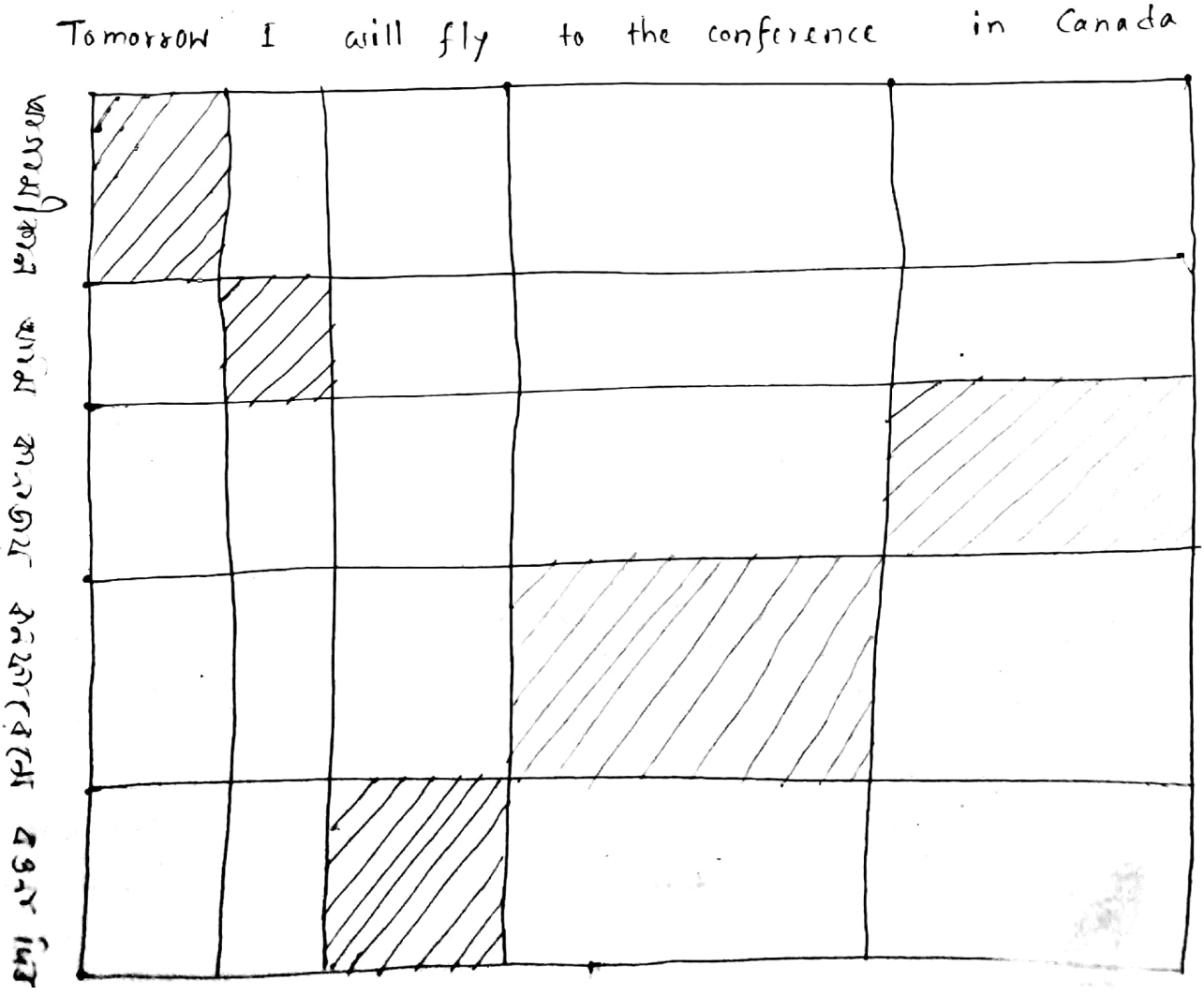
Algorithm:

step1: Foreign input is segmented into phrases
- Any sequence of words

step2: Each phrase is translated into English

step3: phrases are reordered.

phrase alignment:



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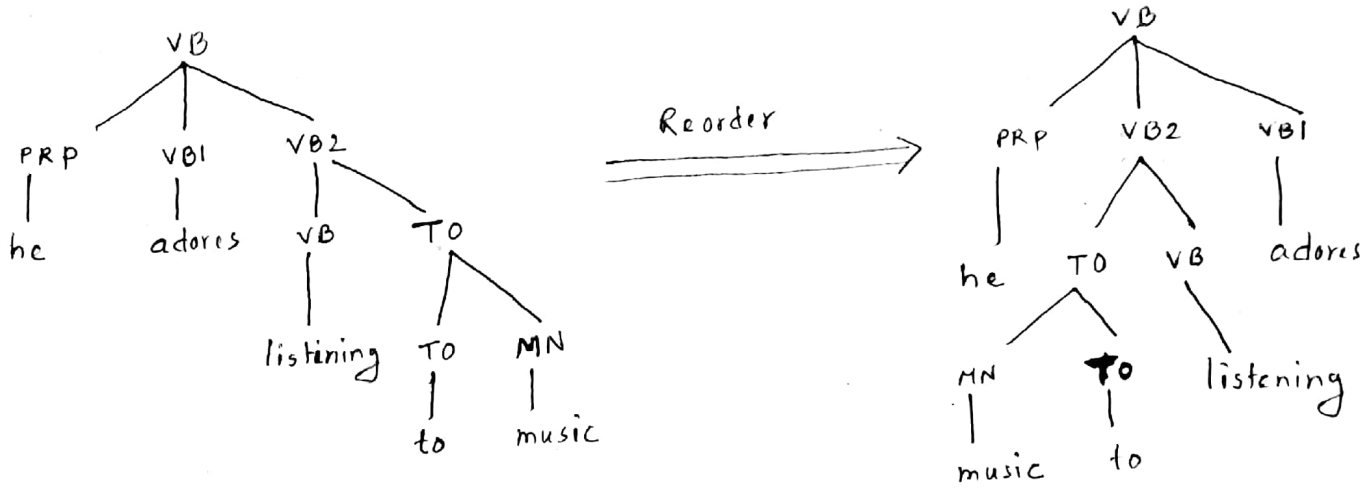
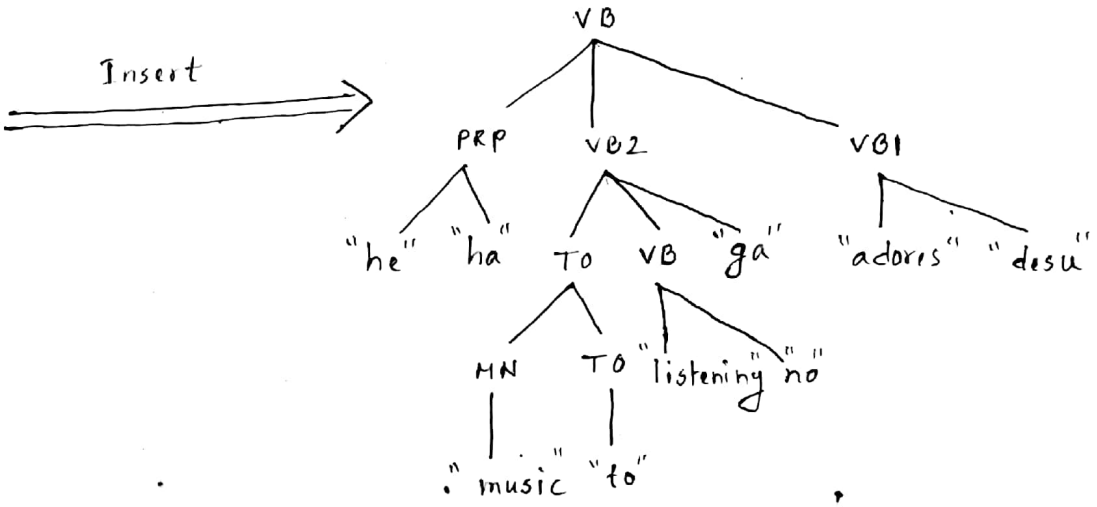
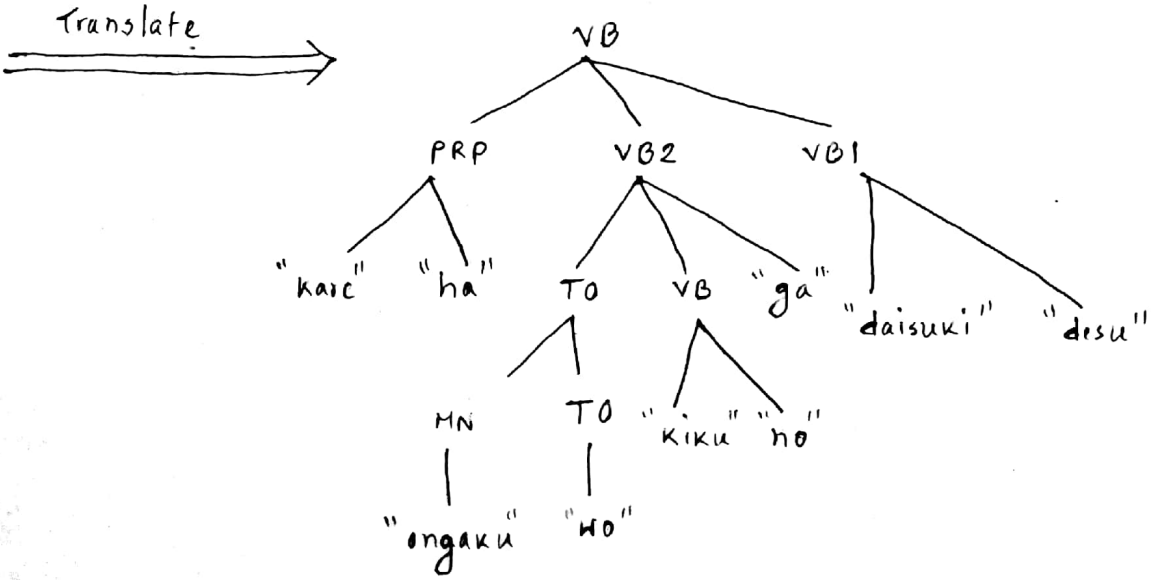
Syntax-based models:







Take leaves

Kare ha ongaku wo kiku no ga daisuki desu

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Performance :

- Word Error Rate
- Word Inaccuracy

Example:

Bangali: তারা মাঠে ফুটবল খেলছে

English: They are playing football in the field

তারা মাঠে ফুটবল খেলছে

Input: তারা | | খেলছে | ফুটবল | | | মাঠে

Output:

They	are	playing	football		the	in field

Total words, $N = 4$
Substitution, $S = 0$
Deletion, $D = 0$
Insertion, $I = 2$

$$\text{Word Error Rate} = 100\% - \text{Word Correct Rate}$$

$$= 100\% - \frac{N - S - D}{N} * 100\%$$

$$= 100\% - \frac{4 - 0 - 0}{4} * 100\%$$

$$= 100\% - 100\%$$

$$= 0\%$$

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IBM Model 1

1. **Generative model:**

Break up translation process into smaller steps.
- IBM Model 1 only uses lexical translation.

2. **Translation probability:**

For a foreign sentence, $f = (f_1, \ldots, f_{l_f})$ of length l_f

To a English sentence, $e = (e_1, \ldots, e_{l_e})$ of length l_e

With an alignment of each english word e_j to a foreign word f_i according to the alignment function $a: j \rightarrow$

$$p(e, a|f) = \frac{\epsilon}{(l_f + 1)^{l_e}} \prod_{j=1}^{l_e} t\left(e_j \mid f_{a(j)}\right)$$

where parameter ϵ is a normalization constant.

Example:

Source sentence, f : das Haus ist klein

Target sentence, e : the house is small

das

e	t(e|f)
the	0.7
that	0.15
which	0.075
who	0.05
this	0.025

Haus

e	t(e|f)
house	0.8
building	0.16
home	0.02
household	0.015
shell	0.005

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ist

e	t(e|f)
is	0.8
's	0.16
exists	0.02
has	0.015
are	0.005

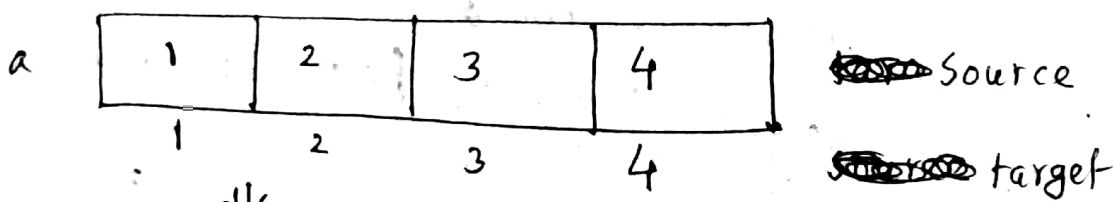
e	t(e|f)
small	0.4
little	0.4
short	0.1
minor	0.06
petty	0.04

Find lexical Translation probability for the given source and target language sentences.

Ans:

$$P(e, a|f) = \frac{\epsilon}{(l_f+1)^{l_e}} \prod_{j=1}^{l_e} t(e_j | f_{a(j)}) \quad \text{..... eqn (1)}$$

	1	2	3	4
Source language sentence, f:	das	Haus	ist	klein
Target language sentence, e:	the	house	is	small
	1	2	3	4



a: {1→1, 2→2, 3→3, 4→4}

Here, $l_f = 4$

$l_e = 4$

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From eq (1)

$$P(e, a|f) = \frac{\epsilon}{(4+1)^4} \prod_{j=1}^{4} t(e_j | f_{a(j)})$$

$$= \frac{\epsilon}{5^4} \cdot \{ t(e_1 | f_{a(1)}) * t(e_2 | f_{a(2)}) * t(e_3 | f_{a(3)}) * t(e_4 | f_{a(4)}) \}$$

$$= \frac{\epsilon}{5^4} \{ t(e_1 | f_1) * t(e_2 | f_2) * t(e_3 | f_3) * t(e_4 | f_4) \}$$

$$= \frac{\epsilon}{5^4} \{ t(\text{the} | \text{das}) * t(\text{house} | \text{Haus}) * t(\text{is} | \text{ist}) * t(\text{small} | \text{klein}) \}$$

$$= \frac{\epsilon}{5^4} (0.7 * 0.8 * 0.8 * 0.4)$$

$$= \frac{0.7 * 0.8 * 0.8 * 0.4}{625} \quad \text{where } \epsilon = 1$$

$$= \frac{0.1792}{625}$$

$$= 0.00028672$$

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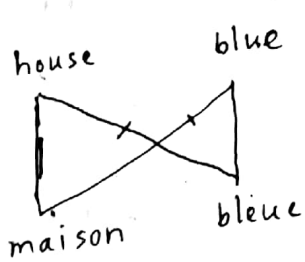
Expection Maximization algorithm :

To estimate translation values from two sentences.



Example:

maison bleue ⟷ blue house

maison ⟷ house

From the example, maison ⟷ blue
maison ⟷ house
bleue ⟷ blue
bleue ⟷ house

From another, maison → house

step1: set parameter values uniformly

$t(\text{bleue}|\text{house}) = \frac{1}{2}$

$t(\text{maison}|\text{house}) = \frac{1}{2}$

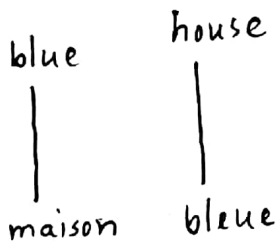
$t(\text{bleue}|\text{blue}) = \frac{1}{2}$

$t(\text{maison}|\text{blue}) = \frac{1}{2}$

Iteration 1:

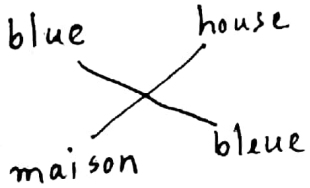
step2: Compute $P(a, f|e)$ for all alignments

Alignment 1:



$P(a,f|e) = t(\text{maison}|\text{blue}) * t(\text{bleue}|\text{ho}$
$= \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$

Alignment 2:



$P(a,f|e) = t(\text{bleue}|\text{blue}) * t(\text{maison}|\text{house})$
$= \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$

Alignment 3:

$P(a,f|e) = t(\text{maison}|\text{house}) = \frac{1}{2}$

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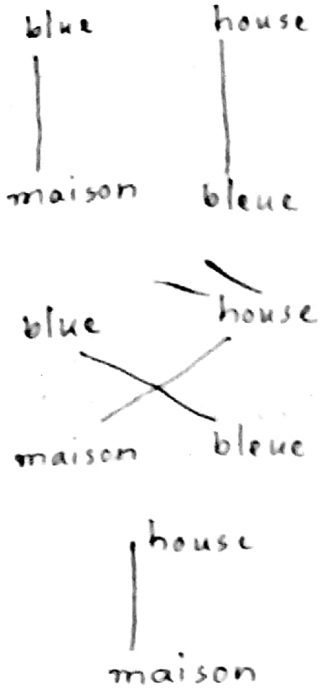
step3: Normalize $P(a, f|e)$ values to yeild $P(a|e,f)$ values



$$P(a|e,f) = \frac{P(a,f|e)}{\sum_a P(a,f|e)} = \frac{\frac{1}{4}}{\frac{1}{4}+\frac{1}{4}} = \frac{\frac{1}{4}}{\frac{2}{4}} = \frac{1}{2}$$

$$P(a|e,f) = \frac{P(a,f|e)}{\sum_a P(a,f|e)} = \frac{\frac{1}{4}}{\frac{1}{4}+\frac{1}{4}} = \frac{1}{2}$$

$$P(a|e,f) = \frac{P(a,f|e)}{\sum_a P(a,f|e)} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$$

[There is only one alignment, so $P(a|e,f)$ will be 1 always]

step4: Collect fractional counts

$tc(\text{maison}|\text{blue}) = \frac{1}{2}$

$tc(\text{bleue}|\text{house}) = \frac{1}{2}$

$tc(\text{bleue}|\text{blue}) = \frac{1}{2}$

$tc(\text{maison}|\text{house}) = \frac{1}{2} + 1 = \frac{3}{2}$

step5: Normalize fractional counts to get revised parameter values

$$t(\text{maison}|\text{blue}) = \frac{\frac{1}{2}}{\frac{1}{2}+\frac{1}{2}} = \frac{1}{2}$$

$$t(\text{bleue}|\text{house}) = \frac{\frac{1}{2}}{\frac{1}{2}+\frac{3}{2}} = \frac{1}{2} * \frac{2}{4} = \frac{1}{4}$$

$$t(\text{bleue}|\text{blue}) = \frac{\frac{1}{2}}{\frac{1}{2}+\frac{1}{2}} = \frac{1}{2}$$

$$t(\text{maison}|\text{house}) = \frac{\frac{3}{2}}{\frac{1}{2}+\frac{3}{2}} = \frac{3}{2} * \frac{2}{4}$$

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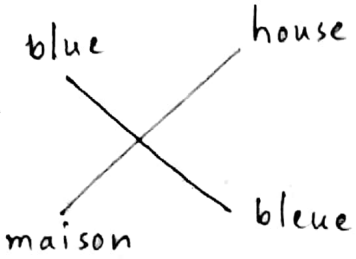
Iteration2 :

step2 : Compute $P(a, f|e)$ for all alignments

Alignment-I

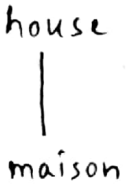
$$P(a, f|e) = t(\text{maison}|\text{blue}) * t(\text{bleue}|\text{house}) = \frac{1}{2} * \frac{1}{4} = \frac{1}{8}$$

Alignment-II



$$P(a, f|e) = t(\text{bleue}|\text{blue}) * t(\text{maison}|\text{house}) = \frac{1}{2} * \frac{3}{4} = \frac{3}{8}$$

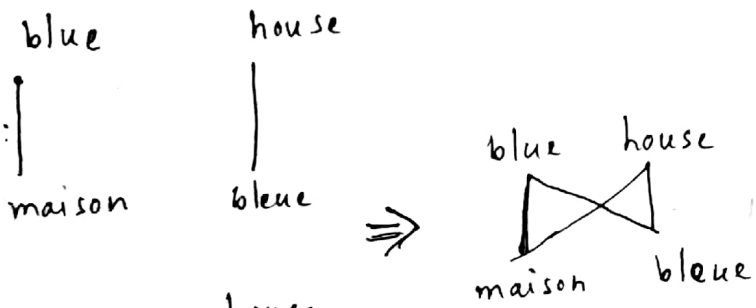
Alignment-III



$$P(a, f|e) = t(\text{maison}|\text{house}) = \frac{3}{4}$$

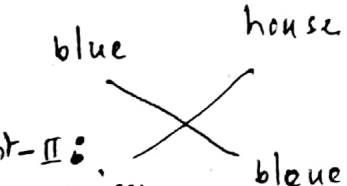
step3 : Normalize $P(a, f|e)$ values to yeild $P(a|e, f)$ values

Alignment-I:



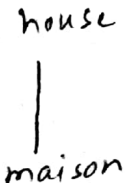
$$P(a|e, f) = \frac{P(a, f|e)}{\sum P(a, f|e)} = \frac{\frac{1}{8}}{\frac{1}{8} + \frac{3}{8}} = \frac{1}{4}$$

Alignment-II :



$$P(a|e, f) = \frac{P(a, f|e)}{\sum P(a, f|e)} = \frac{\frac{3}{8}}{\frac{1}{8} + \frac{3}{8}} = \frac{3}{4}$$

Alignment-III :



$$P(a|e, f) = \frac{P(a, f|e)}{\sum P(a, f|e)} = \frac{\frac{3}{4}}{\frac{3}{4}} = 1$$

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house)

house)

step4: Collect fractional counts

$tc(\text{maison} \mid \text{blue}) = \frac{1}{4}$

$tc(\text{bleue} \mid \text{house}) = \frac{1}{4}$

$tc(\text{bleue} \mid \text{blue}) = \frac{3}{4}$

$tc(\text{maison} \mid \text{house}) = \frac{3}{4} + 1 = \frac{7}{4}$

step5: Normalize fractional counts to get revised parameter values

$$t(\text{maison} \mid \text{blue}) = \frac{\frac{1}{4}}{\sum tc(\ldots \mid \text{blue})}$$

$$= \frac{\frac{1}{4}}{\frac{1}{4} + \frac{3}{4}} = \frac{1}{4}$$

$$t(\text{bleue} \mid \text{house}) = \frac{\frac{1}{4}}{\sum tc(\ldots \mid \text{house})}$$

$$= \frac{\frac{1}{4}}{\frac{1}{4} + \frac{7}{4}}$$

$$= \frac{1}{8}$$

$$t(\text{bleue} \mid \text{blue}) = \frac{\frac{3}{4}}{\sum tc(\ldots \mid \text{blue})} = \frac{\frac{3}{4}}{\frac{1}{4} + \frac{3}{4}} = \frac{3}{4}$$

$$t(\text{maison} \mid \text{house}) = \frac{\frac{7}{4}}{\frac{1}{4} + \frac{7}{4}} = \frac{\frac{7}{4}}{\frac{8}{4}} = \frac{7}{8}$$

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Repeating steps 2-5 many times yields

$t(\text{bleue} \mid \text{house}) = 0.0001$

$t(\text{maison} \mid \text{house}) = 0.9999$ ✓

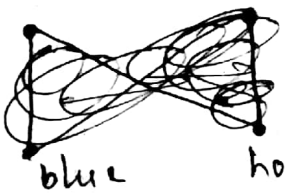
$t(\text{bleue} \mid \text{blue}) = 0.9999$ ✓

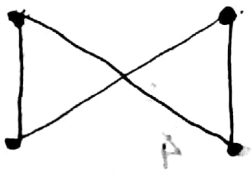
$t(\text{maison} \mid \text{blue}) = 0.0001$

maison bleue ⟷ blue house

maison ⟷ house

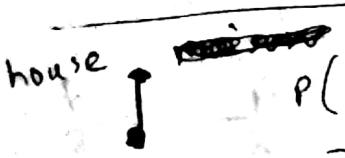
So we can calculate translation probabilites by the way of these alignment probabilites





$$P(\underline{f} \mid \underline{e}) = \sum_{a} P(a, f \mid e)$$

$$= \sum_{a} \prod_{j=1}^{m} t(f_j \mid e_i)$$



$P(f \mid e)$

$= \sum_{a} P(a, f \mid e)$

$= t(\text{maison} \mid \text{blue}) * t(\text{blue} \mid \text{house})$

$+ t(\text{blue} \mid \text{blue}) * t(\text{maison}$

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Dr. Mohammad Nurul Huda

CSE 6015, Statistical Machine Translation

Lecture sheet-7

Computing Model 1 parameters:

How to compute translation probabilities for model 1 from a parallel corpus?

Step1: Determine candidates. For each English word e collect all foreign words f that co-occur at least once with e.

Step2: Initialize $P(f|e)$ uniformly

$$P(f|e) = \frac{1}{\text{No of co-occuring foreign words}}$$

Step3: Iteratively refine translation probabilities

```
for n iterations
    set tc to zero
    for each sentence pair (e, f) of lengths (l, m)
        for j = 1 to m
            total = 0
            for i = 1 to l
                total = total + P(f_j | e_i)
            end for
            for i = 1 to l
                tc(f_j | e_i) = tc(f_j | e_i) + P(f_j | e_i) / total
            end for
        end for
    end for
    for each word e
        total = 0
        for each word f
            total = total + tc(f|e)
        end for
        for each word f
            P(f|e) = tc(f|e) / total
        end for
    end for
end for
```

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Example: (parallel corpus)

the dog :: le chien

the cat :: le chat

step 1+2 (collect candidates and initialize uniformly)

the → le, chien, chat

dog → le, chien, chat

cat → le, chat, chien

NULL → le, chien, chat

$$P(le|the) = P(chien|the) = P(chat|the) = \frac{1}{3}$$

$$P(le|dog) = P(chien|dog) = P(chat|dog) = \frac{1}{3}$$

$$P(le|cat) = P(chat|cat) = P(chien|cat) = \frac{1}{3}$$

$$P(le|NULL) = P(chien|NULL) = P(chat|NULL) = \frac{1}{3}$$

step 3: Iterate

e_1 NULL e_2 the e_3 dog :: f_1 le f_2 chien

e: $l = 3$ f: $m = 2$

$j = 1$ total = 0 $i = 1$ total = $0 + P(f_1|e_1)$

$i = 2$ total = $P(f_1|e_1) + P(f_1|e_2)$

$i = 3$ total = $P(f_1|e_1) + P(f_1|e_2) + P(f_1|e_3)$

$= P(le|NULL) + P(le|the) + P(le|dog)$

$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$

$i = 1$ $tc(f_1|e_1) = 0 + \frac{P(f_1|e_1)}{total} =$

$\Rightarrow tc(le|NULL) = P(le|NULL)/1 = \frac{1/3}{1} = 0.33$

$i = 2$ $tc(le|the) = P(le|the)/1 = \frac{1/3}{1} = 0.333$

$i = 3$ $tc(le|dog) = P(le|dog)/1 = \frac{1/3}{1} = 0.333$

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$j = 2$ total = 0 $i = 1$ total = $0 + P(f_2|e_1) = P(f_1|e_1)$

$i = 2$ total = $P(f_2|e_1) + P(f_2|e_2)$

$i = 3$ total = $P(f_2|e_1) + P(f_2|e_2) + P(f_2|e_3)$

$= P(\text{chien}|\text{NULL}) + P(\text{chien}|\text{the}) + P(\text{chien}|\text{dog})$

$= 1$

$i = 1$ $tc(f_2|e_1) = 0 + \frac{P(f_2|e_1)}{\text{Total}} = 0 + \frac{\frac{1}{3}}{1} = 0.33$

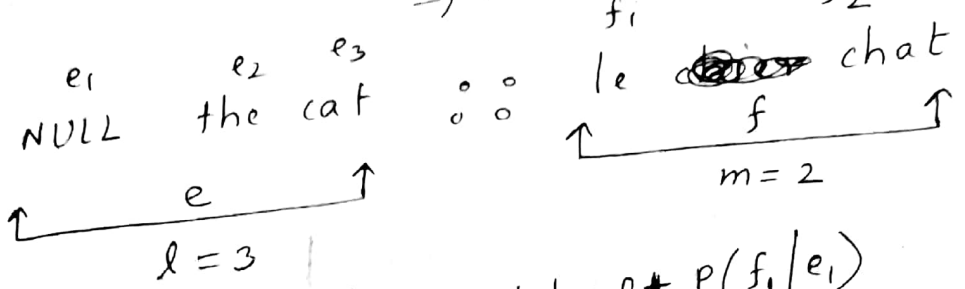
$\Rightarrow tc(\text{chien}|\text{NULL})$

$i = 2$ $tc(f_2|e_2) = 0 + \frac{P(f_2|e_2)}{\text{Total}} = 0 + \frac{\frac{1}{3}}{1} = 0.33$

$\Rightarrow tc(\text{chien}|\text{the})$

$i = 3$ $tc(f_2|e_3) = 0 + \frac{P(f_2|e_3)}{\text{Total}} = 0 + \frac{\frac{1}{3}}{1} = 0.33$

$\Rightarrow tc(\text{chien}|\text{dog})$



$j = 1$ total = 0 $i = 1$ total = $0 + P(f_1|e_1)$

$i = 2$ total = $P(f_1|e_1) + P(f_1|e_2)$

$i = 3$ total = $P(f_1|e_1) + P(f_1|e_2) + P(f_1|e_3)$

$= P(\text{le}|\text{NULL}) + P(\text{le}|\text{the}) + P(\text{le}|\text{cat})$

$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$

$i = 1$ $tc(\text{le}|\text{NULL}) = 0.333 + \frac{P(\text{le}|\text{NULL})}{\text{total}} = 0.333 + 0.333 = 0.666$

$i = 2$ $tc(\text{le}|\text{the}) = 0.333 + \frac{P(\text{le}|\text{the})}{\text{total}} = 0.333 + 0.333 = 0.666$

$i = 3$ $tc(\text{le}|\text{cat}) = 0 + \frac{P(\text{le}|\text{cat})}{\text{total}} = 0 + 0.333 = 0.333$

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$j=2$ total $=0$ $i=1$

$i=2$ total $= P(f_2|e_1) + P(f_2|e_2)$

$i=3$ total $= P(f_2|e_1) + P(f_2|e_2) + P(f_2|e_3)$

$= P(\text{chat}|\text{NULL}) + P(\text{chat}|\text{the}) + P(\text{chat}|\text{cat})$

$= 1$

$i=1$ $tc(\text{chat}|\text{NULL}) = 0 + \frac{P(\text{chat}|\text{NULL})}{\text{total}} = 0 + \frac{1/3}{1} = 0.333$

$i=2$ $tc(\text{chat}|\text{the}) = 0 + \frac{P(\text{chat}|\text{the})}{\text{total}} = 0 + \frac{1/3}{1} = 0.333$

$i=3$ $tc(\text{chat}|\text{cat}) = 0 + \frac{P(\text{chat}|\text{cat})}{\text{total}} = 0 + \frac{1/3}{1} = 0.333$

(Recompute translation probabilities)

$\text{total}(\text{the}) = 0 + tc(\text{le}|\text{the}) + tc(\text{chien}|\text{the}) + tc(\text{chat}|\text{the})$

$= 0.666 + 0.333 + 0.333$

$= 1.333$

$P(\text{le}|\text{the}) = \frac{tc(\text{le}|\text{the})}{\text{total}(\text{the})} = \frac{0.666}{1.333} = 0.5$

$P(\text{chien}|\text{the}) = \frac{tc(\text{chien}|\text{the})}{\text{total}(\text{the})} = \frac{0.333}{1.333} = 0.25$

$P(\text{chat}|\text{the}) = \frac{tc(\text{chat}|\text{the})}{\text{total}(\text{the})} = \frac{0.333}{1.333} = 0.25$

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total (dog):

$$total(dog) = 0 + tc(le|dog) + tc(chien|dog)$$
$$= 0 + \frac{1}{3} + \frac{1}{3}$$
$$= 0.666$$

$$P(le|dog) = tc(le|dog) / total(dog)$$
$$= \frac{0.333}{0.666}$$
$$= 0.5$$

$$P(chien|dog) = tc(chien|dog) / total(dog)$$
$$= \frac{0.333}{0.666}$$
$$= 0.5$$

total (cat):

$$total(cat) = 0 + tc(le|cat) + tc(chien|cat)$$
$$= 0.333 + 0.333$$
$$= 0.666$$

$$P(le|cat) = \frac{tc(le|cat)}{total(cat)}$$
$$= \frac{0.333}{0.666}$$
$$= 0.5$$

$$P(chat|cat) = \frac{tc(chat|cat)}{total(cat)}$$
$$= \frac{0.333}{0.666}$$
$$= 0.5$$

total (NULL):

$$total(NULL) = 0 + tc(le|NULL) + tc(chien|NULL) + tc(chat|NULL)$$
$$= 0.666 + 0.333 + 0.333 = 1.332$$

$$P(le|NULL) = \frac{tc(le|NULL)}{total(NULL)} = \frac{0.666}{1.332}$$

$$P(chien|NULL) = \frac{tc(chien|NULL)}{total(NULL)} = \frac{0.333}{1.332}$$

$$P(chat|NULL) = \frac{tc(chat|NULL)}{total(NULL)} = \frac{0.333}{1.332}$$

iteration 2:

Go to step 3

iteration 3:

Go to step 3

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